

# An Entropic Perspective on P vs NP: Reformulation via Kolmogorov Complexity

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We present a unified framework for the well-known equivalence between the P vs NP problem and resource-bounded Kolmogorov complexity, framing it as an *entropic reformulation*:  $P \neq NP$  is equivalent to the absence of a polynomial-time witness selector whose outputs have only logarithmic resource-bounded conditional complexity. This equivalence has been known in various forms since the work of Orponen et al. (1994) and Fortnow (2000). This paper does not prove  $P \neq NP$ ; rather, it provides a unified information-theoretic presentation that identifies the precise structure of what must be shown and why classical barrier techniques are not obstructions to this approach. We introduce the concept of *algorithmic positivity* – an NP language is algorithmically positive if it admits a polynomially scaling witness-producing function whose output has low resource-bounded conditional Kolmogorov complexity  $K^t(W(x) \mid x) \leq O(\log |x|)$  – and characterize the *Entropic Separation Conjecture* (no NP-complete language is algorithmically positive) as *equivalent* to  $P \neq NP$ , establishing ESC as a reformulation of the P vs NP problem in information-theoretic language rather than an independent result. We conduct a systematic barrier analysis, arguing that arguments based on unbounded Kolmogorov complexity  $K$  are not naturally subject to the three classical barriers (relativization, natural proofs, algebrization), while noting that the barrier status for the resource-bounded variant  $K^t$  used in our core results remains to be fully established. The central characterization is the *witness entropy gap*:  $P = NP$  would imply the existence of a canonical polynomial-time witness selector with  $K^t(W(x) \mid x) = O(\log |x|)$ , while  $P \neq NP$  implies that, for every fixed polynomial time bound and logarithmic constant, infinitely many instances admit no valid witness with such a short resource-bounded conditional description. The stronger near-maximal bound  $K^t(w \mid x) \geq |w| - O(\log n)$  is treated below as an additional bridge hypothesis, not as a consequence of  $P \neq NP$ . We identify the precise remaining challenge: converting this characterization into a proof requires a *uniformity bridge* – a new connection between algorithmic information theory and uniform complexity theory.

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## I. INTRODUCTION

### A. The problem

The P vs NP question asks whether every problem whose solutions can be *verified* in polynomial time can also be *solved* in polynomial time. Formally:

**Definition I.1.** P is the class of languages decidable in polynomial time. NP is the class of languages for which membership has a polynomial-time verifiable witness [11].  $P = NP$  iff every language in NP is also in P. By the Cook–Levin theorem [20], this is equivalent to the satisfiability problem (SAT) being solvable in polynomial time.

Despite decades of effort, the problem remains open. This paper argues that the difficulty is not technical but *conceptual*: previous approaches attempt to prove lower bounds via *constructive* methods (exhibiting specific hard instances, explicit circuit separations, or algebraic obstructions), which are systematically blocked by three barriers. We propose an alternative: reformulate  $P \neq NP$  via a *structural entropy argument* that identifies the impossibility of witness compression as the fundamental obstruction, and pinpoint the precise remaining gap that any proof must close.

### B. The three barriers

Three meta-theorems limit the scope of known proof techniques:

**1. Relativization** [1]. There exist oracles  $A, B$  with  $P^A = NP^A$  and  $P^B \neq NP^B$ . Therefore, any proof of  $P \neq NP$  must use techniques that do not relativize – it must exploit the internal structure of computation, not just input-output behavior.

**2. Natural proofs** [2]. If strong pseudorandom generators exist (a widely believed assumption), then no “natural” proof method – one that is both *constructive* (recognizes hard functions efficiently) and *large* (applies to a dense set of functions) – can prove superpolynomial circuit lower bounds against  $P/\text{poly}$ .

**3. Algebrization** [3]. Even non-relativizing techniques based on arithmetization (as in  $IP = PSPACE$ ) cannot separate P from NP, because there exist algebraic extensions of oracles in which both  $P = NP$  and  $P \neq NP$  hold.

### C. The positivity alternative

We observe a *superficial structural parallel* across several major mathematical problems: breakthroughs often succeed not by constructing the desired object, but by showing its non-existence would violate a *positivity condition*. The analogy is at the level of proof strategy (non-constructive arguments via non-negative quantities),

not at the level of mathematical content – the positivity structures involved are fundamentally different in nature (see Table I):

Table I. Positive structures and proof strategies across foundational mathematical and computational problems.

| Problem                      | Positive structure           | Status            |
|------------------------------|------------------------------|-------------------|
| Weil conjectures             | Frobenius eigenvalues        | Proven            |
| BSD ( $\text{rank} \leq 1$ ) | Néron-Tate height            | Proven            |
| Hodge conjecture             | Hodge–Riemann form           | Open              |
| RH                           | Spectral zeta flow           | [17]              |
| <b>P vs NP</b>               | <b>Kolmogorov complexity</b> | <b>This paper</b> |
| Broader program              | Free energy framework        | [16]              |

*Caveat:* This table illustrates structural parallels at the level of proof strategy; it does not claim that  $P \neq NP$  is solved or simplified by the positivity perspective. The positivity structures involved are fundamentally different in each case.

The organizing observation: *Kolmogorov complexity serves as a natural non-negative quantity for computational complexity*. It is non-negative ( $K(x) \geq 0$ ), machine-invariant (up to additive constants), and fundamentally non-constructive (not computable) – properties that allow unbounded  $K$  to bypass the three barriers. The resource-bounded variant  $K^t$ , which is essential for the core results, inherits some but not all of these properties; its barrier status is discussed in Section III.

This paper is a **presentation and framework paper**: it presents the known equivalence between  $P \neq NP$  and an entropic condition in a unified framework (structured by the claim-evidence-gate ledger in Table II), but does not prove the separation itself and does not claim the equivalence as a new result (see Related Work, Section X, for the priority of Fortnow [24] and Orponen et al. [25]). The paper is self-contained; all main results depend only on definitions and proofs within this document. Connections to the broader FST program are cited for context but are not logically required.

## II. SYSTEMATIC BARRIER ANALYSIS

### A. Why constructive methods fail

All three barriers share a common cause: they block proofs that treat computational problems as *black boxes* or that rely on *statistical* properties of Boolean functions.

- **Relativization** blocks proofs that treat the computation as input-output behavior (oracle access), ignoring internal structure.
- **Natural proofs** block proofs that identify hard functions via efficiently testable, statistically dense properties.

Table II. Claim-evidence-gate ledger for the paper.

| Layer                       | Visible evidence in the paper                               | Permitted reading                      | Stop gate before stronger claims           |
|-----------------------------|-------------------------------------------------------------|----------------------------------------|--------------------------------------------|
| Entropic equivalence        | ESC/EH propositions and remarks, Secs. V–VII                | Known-equivalence framework            | EH or Uniformity Bridge proof              |
| Barrier analysis            | S1–S5 and $K$ versus $K^t$ caveat                           | $K$ motivates; $K^t$ remains hard case | No resource-bounded bypass claim           |
| Restricted-model guardrails | OGP, low-degree, stability, AC0, MCSP                       | Restricted-front obstruction only      | Map uniform search to exact front class    |
| Illustrative numerics       | SAT entropy diagnostics                                     | Exploratory stress test                | Asymptotic family, null model, denominator |
| Open bridge program         | UB, advice erosion, flow, capillary, Williams slice entropy | Lemma targets                          | Theorem: search implies entropy collapse   |

- **Algebrization** blocks proofs that use only the polynomial structure of arithmetized computations.

The common thread: all three barriers apply to *model-dependent* arguments – arguments that work within a specific computational model (oracle machines, circuits, algebraic extensions) rather than targeting the *intrinsic information content* of the computation.

### B. The specification for a valid proof

From the barrier analysis, we derive necessary conditions for any proof of  $P \neq NP$ :

- (S1) **Non-relativizing:** The argument must exploit internal computational structure, not just oracle behavior.
- (S2) **Non-natural:** The argument must not define a “largeness” property that is efficiently computable. It must target individual objects or non-constructive properties.
- (S3) **Non-algebrizing:** The argument must go beyond polynomial identity testing and arithmetization.
- (S4) **Model-independent:** The argument must not depend on a specific computational model (circuits, Turing machines, etc.) but on an invariant structural property.
- (S5) **Robust:** The argument must not break when moving from restricted models (monotone circuits,  $AC^0$ ) to general computation.

We summarize the five barrier specifications and their corresponding entropic bypass mechanisms in Table III. We now show that algorithmic entropy satisfies all five specifications.

## III. ALGORITHMIC ENTROPY AS POSITIVITY STRUCTURE

### A. Kolmogorov complexity

Let  $U$  be a fixed universal prefix-free Turing machine. The *Kolmogorov complexity* (algorithmic entropy) of a finite binary string  $x$  is:

$$K(x) := \min\{|p| : U(p) = x\}. \quad (1)$$

The *conditional* Kolmogorov complexity of  $x$  given  $y$  is:

$$K(x | y) := \min\{|p| : U(p, y) = x\}. \quad (2)$$

**Theorem III.1** (Invariance [12, 13, 22]). *For any two universal prefix-free machines  $U, U'$ :*

$$|K_U(x) - K_{U'}(x)| \leq c_{U, U'}, \quad (3)$$

where  $c_{U, U'}$  is a constant independent of  $x$ .

**Theorem III.2** (Incomputability).  *$K(x)$  is not computable. More precisely: for every total computable function  $f$ , there exists  $x$  with  $K(x) > f(x)$ .*

**Theorem III.3** (Incompressibility). *For every  $n$ , at least  $2^n - 2^{n-c+1} + 1$  strings of length  $n$  satisfy  $K(x) \geq n - c$ .*

**Theorem III.4** (Symmetry of Information [4]). *For all strings  $x, y$ :*

$$K(x, y) = K(x) + K(y | x, K(x)) + O(\log(K(x, y))). \quad (4)$$

*This implies that conditional complexity is well-behaved: the information in the pair  $(x, y)$  decomposes additively (up to logarithmic terms) into the information in  $x$  plus the additional information in  $y$  given  $x$ .*

### B. Why unbounded Kolmogorov complexity bypasses the barriers

**Critical scope limitation.** The barrier immunity analysis below applies to *unbounded* Kolmogorov complexity  $K$ , which is not the quantity used in the core results of this paper. The core results (Theorem IV.3, the witness entropy gap) use *resource-bounded*  $K^t$ , for which barrier immunity is weaker and partially open. This section therefore provides *motivation* for the entropy approach but does *not* establish barrier immunity for the paper’s actual technical results. The reader should consult Remark III.6 for the precise status of  $K^t$ .

**Claim III.5** (Barrier immunity). Arguments based on *unbounded* Kolmogorov complexity  $K$  plausibly satisfy specifications (S1)–(S5). We present heuristic arguments for each; a fully rigorous treatment is an open research question (see Limitations, Section XI). **Important caveat:** The core technical results of this paper use *resource-bounded*  $K^t$ , for which barrier immunity is weaker and partially open – see Remark III.6 below.

Table III. Synoptic barrier analysis and Kolmogorov complexity bypass mechanism.

| Barrier          | Foundational Reference         | Blocked Proof Mechanism                      | Specification          | Kolmogorov Complexity Bypass Mechanism                                                |
|------------------|--------------------------------|----------------------------------------------|------------------------|---------------------------------------------------------------------------------------|
| Relativization   | Baker–Gill–Solovay (1975) [1]  | Input-output oracle simulations              | (S1) Non-relativizing  | Defined over fixed machine; intrinsic information; robust under polynomial slowdowns  |
| Natural Proofs   | Razborov–Rudich (1997) [2]     | Efficiently testable, dense properties       | (S2) Non-natural       | Non-computability of $K(x)$ prevents polynomial-time largeness testing                |
| Algebrization    | Aaronson–Wigderson (2009) [3]  | Low-degree polynomial extensions             | (S3) Non-algebrizing   | Minimization of program length invariant under algebraic oracle lifts                 |
| Model Dependence | Classical circuit bounds       | Gates/depth-specific combinatorial arguments | (S4) Model-independent | Universal invariance theorem up to machine-dependent constants                        |
| Non-Robustness   | Restriction-specific collapses | Methods failing outside $AC^0$ /monotone     | (S5) Robust            | Information-theoretic gap $\Omega( w )$ vs $O(\log n)$ independent of gate primitives |

**(S1) Non-relativizing.**  $K(x)$  is defined without reference to any oracle – it is an absolute, fixed quantity for each string  $x$ . The witness entropy gap (Theorem IV.3) uses  $K^t(w | x)$  (resource-bounded conditional complexity). We note an important subtlety: the invariance theorem for resource-bounded  $K^t$  is weaker than for unbounded  $K$  – changing the universal machine can introduce a *polynomial* slowdown factor rather than just an additive constant [4]. However, the argument remains non-relativizing because  $K^t$  is defined over a fixed computational model without oracle access. A relativizing proof would need to hold for all oracles  $A$ , but our argument targets oracle-independent quantities: the gap between  $O(\log n)$  and  $\Omega(|w|)$  is superpolynomial and thus robust under polynomial slowdown factors.

**(S2) Non-natural.**  $K(x)$  is not computable (Theorem III.2), so no efficient procedure can test whether a specific function has “high Kolmogorov complexity.” This directly violates the “constructiveness” requirement of natural proofs.

**(S3) Non-algebrizing.** Algebrization applies to proof techniques that use only the polynomial structure of arithmetized computations. Arguments based on  $K^t$  operate over the full power of Turing machine computation, not polynomial identity testing. While it remains a research question whether  $K^t$ -based arguments are formally non-algebrizing in the technical sense of [3], the entropy framework is not naturally expressible as an algebraic identity, suggesting it escapes this barrier.

**(S4) Model-independent.** The invariance theorem ensures that  $K(x)$  is the same (up to  $O(1)$ ) for all universal machines – Turing machines, circuits,  $\lambda$ -calculus, etc.

**(S5) Robust.**  $K(x)$  does not depend on the computational model used to define it.

*Remark III.6* (Barrier immunity: unbounded  $K$  vs. resource-bounded  $K^t$ ). The barrier immunity claimed above applies to unbounded Kolmogorov complexity  $K$ , which is an absolute, uncomputable quantity invariant up to  $O(1)$  under change of universal machine. However, the core technical results of this paper (Theorem IV.3, the witness entropy gap) necessarily use *resource-bounded*  $K^t$ , which is computable and whose invariance is weaker (polynomial slowdown rather than additive constant). The barrier status of  $K^t$ -based arguments is therefore less clear-cut:

- **Relativization:**  $K^t(w | x)$  is defined without oracles, so our arguments are non-relativizing in the sense that they do not hold “for all oracles.” However,  $K^{t,A}(w | x)$  (with oracle access) can differ substantially from  $K^t(w | x)$ , and a full barrier-immunity proof for the resource-bounded case remains open.
- **Natural proofs:**  $K^t$  is computable, so unlike  $K$ , it does not automatically escape the constructiveness requirement. The argument that our approach avoids natural proofs rests on the fact that we do not use  $K^t$  as an efficiently testable “largeness” property of Boolean functions, but this distinction deserves further formalization.
- **Algebrization:** No formal proof exists that  $K^t$ -based arguments are non-algebrizing in the technical sense of [3]. S3 (non-algebrization for  $K^t$ ) remains an open question; we make no claim that our framework provably evades this barrier.

In summary: the barrier bypass is rigorous for unbounded  $K$  but remains heuristic for resource-bounded  $K^t$ . Since the witness entropy gap requires  $K^t$ , establishing formal barrier immunity for the resource-bounded case is an important open problem.

## IV. THE WITNESS ENTROPY GAP

### A. NP witnesses as compressed information

For an NP language  $L$ , every yes-instance  $x \in L$  has a polynomial-length witness  $w$  verifiable in polynomial time:

$$x \in L \iff \exists w, |w| \leq \text{poly}(|x|), V(x, w) = 1. \quad (5)$$

The conditional complexity of the witness given the instance measures how much *additional* information the witness contains beyond the instance:

$$K(w | x) = \begin{array}{l} \text{extra information needed} \\ \text{to produce } w \text{ from } x. \end{array} \quad (6)$$

### B. What $P = NP$ would imply

**Proposition IV.1** (Witness compression under  $P = NP$ ). *If  $P = NP$ , then for every fixed polynomially balanced NP verifier relation for  $L$  and every  $x \in L$ , there exists a*

polynomial-time canonical witness selector  $A$  that produces a valid witness  $w$  from  $x$ :

$$w = A(x). \quad (7)$$

Therefore, for some polynomial  $q$  depending on the running time of  $A$ :

$$K^{q(|x|)}(w \mid x) \leq K(A) + O(\log |x|) = O(\log |x|), \quad (8)$$

since the algorithm  $A$  is a fixed finite program running in polynomial time, and  $O(\log |x|)$  bits encode the input length.

*Proof.* If  $P = NP$ , the search version of SAT is solvable in polynomial time (by self-reducibility). Given a satisfiable formula  $\varphi$ , the algorithm  $A$  finds a satisfying assignment bit-by-bit in polynomial time. For a fixed NP verifier relation, the same prefix-extension test is an NP predicate and is therefore decidable in polynomial time under  $P = NP$ . The program “run  $A$  on input  $x$ ” has length  $|A| + O(\log |x|)$ , where  $|A|$  is a constant.  $\square$

### C. The entropy obstruction

*Remark IV.2* (Unbounded vs. resource-bounded complexity – why  $K^t$  is essential). The distinction between  $K(w \mid x)$  and  $K^t(w \mid x)$  is crucial and must be understood before the main theorem. For any decidable language  $L$  and any  $(x, w)$  with  $V(x, w) = 1$ , the brute-force verifier-search has  $K(w \mid x) \leq O(1)$  (the search program has constant length – it simply enumerates all candidates and checks each). The information-theoretic content of NP-hardness lies entirely in the *time* required to extract the witness, not in the *description length*. This is why  $K^t$  with polynomial  $t$  is the correct measure: it captures the computational cost of witness extraction, which is the essence of the P vs NP question.

**Theorem IV.3** (Superlogarithmic witness obstruction – resource-bounded, conditional (cf. Fortnow [24], Allender [23])). *The following is a reformulation of known connections between resource-bounded Kolmogorov complexity and P vs NP (see Related Work, Section X) in the specific language of conditional witness entropy.*

*If  $P \neq NP$ , then for SAT, for every polynomial  $q$  and every constant  $C > 0$ , there exist infinitely many satisfiable formulas  $x$  such that **every** satisfying assignment  $w$  satisfies:*

$$K^{q(|x|)}(w \mid x) > C \log |x|. \quad (9)$$

*That is, for every fixed polynomial time bound and every fixed logarithmic advice constant, infinitely many instances have no valid witness with that small a time-bounded conditional description. The same statement holds for any NP-complete search problem to which SAT reduces by a polynomial-time witness-preserving search reduction.*

*Equivalently (by contraposition): if there exist a polynomial  $q$  and a constant  $C$  such that every satisfiable formula  $x$  has some satisfying assignment  $w$  with  $K^{q(|x|)}(w \mid x) \leq C \log |x|$ , then  $P = NP$ .*

*Note on quantifier order: this theorem is deliberately stated for each fixed pair  $(q, C)$ . It does not assert a single infinite instance family that resists all polynomial time bounds simultaneously, and it does not assert near-maximal entropy of witnesses.*

*Proof.* We prove the contrapositive for SAT. Suppose some polynomial  $q$  and constant  $C$  have the property that every satisfiable formula  $x$  has a satisfying assignment  $w$  with  $K^{q(|x|)}(w \mid x) \leq C \log |x|$ . Then for every satisfiable  $x$ , there exists a program  $p_x$  of length at most  $C \log |x|$  that produces a satisfying assignment within  $q(|x|)$  steps. There are at most  $|x|^C$  such programs. A polynomial-time SAT search algorithm can enumerate all programs of length at most  $C \log |x|$ , run each for  $q(|x|)$  steps on input  $x$ , and verify each output. Since SAT search is polynomial-time equivalent to SAT decision by self-reducibility, this would imply  $P = NP$ .

The “infinitely many” clause follows by the same contrapositive plus finite hardcoding: if only finitely many formulas failed the bound, their witnesses could be stored in a finite lookup table and the above enumeration would solve all remaining instances in polynomial time. Constructing such hard instances explicitly is not provided by the argument; the theorem is a non-constructive reformulation.

*Why unbounded  $K$  fails:* For any SAT instance  $x$  with a satisfying assignment  $w$ , the brute-force program “enumerate all assignments, test each, output the first satisfying one” has length  $O(1)$  and finds  $w$  from  $x$  in time  $O(2^{|x|})$ . Thus  $K(w \mid x) \leq O(1)$  – the unbounded conditional complexity is trivially small. This is why the resource bound is essential: the information-theoretic content of NP-hardness lies in the *time* required to extract the witness, not in the description length.  $\square$

**Conjecture IV.4** (Near-maximal witness entropy). *For some NP-complete search formulation, and for every polynomial  $q$ , there exist infinitely many yes-instances  $x$  such that every valid witness  $w$  satisfies*

$$K^{q(|x|)}(w \mid x) \geq |w| - O(\log |x|). \quad (10)$$

*This near-maximal strengthening would yield the sharp  $\Omega(|w|)$  vs.  $O(\log |x|)$  entropy gap used heuristically in several bridge targets below. It is not implied by Theorem IV.3; it is an additional open hypothesis.*

### D. The gap

Combining Proposition IV.1 and Theorem IV.3:

**Corollary IV.5** (Witness Entropy Gap – the core contradiction structure). *For any fixed polynomial time bound*

$q$  and logarithmic constant  $C$ , Theorem IV.3 gives infinitely many SAT instances whose valid witnesses are not  $C \log |x|$ -compressible within time  $q(|x|)$ . For unique-witness instances, the contradiction structure is sharpest:

- $P = NP \Rightarrow$  a polynomial-time canonical witness selector  $A$  exists with  $K^{q(|x|)}(A(x) \mid x) \leq O(\log |x|)$  (Proposition IV.1).
- $P \neq NP \Rightarrow$  for every fixed  $(q, C)$ , infinitely many instances have no valid witness  $w$  with  $K^{q(|x|)}(w \mid x) \leq C \log |x|$  (Theorem IV.3).

This is not a proof of  $P \neq NP$ , but a characterization of the logarithmic-compression threshold:  $P = NP$  if and only if a polynomial-time canonical witness selector with logarithmic conditional description exists for SAT. The stronger  $\Omega(|w|)$  vs.  $O(\log n)$  gap follows only under the near-maximal witness entropy hypothesis (Conjecture IV.4).

The gap is sharpest for instances with a unique satisfying assignment; for instances with multiple witnesses, the theorem rules out any logarithmically short polynomial-time witness description on the hard instances.

**Remark IV.6** (Unique vs. multiple witnesses). Theorem IV.3 guarantees instances where every valid witness has superlogarithmic conditional complexity relative to the fixed time bound and constant under consideration. For these instances, the contradiction structure holds regardless of how many witnesses exist: any algorithm  $A$  must output some valid witness, and none of the valid witnesses has the prohibited logarithmically short description.

For unique-witness instances, the structure is especially clean:  $A$  has no choice over which witness to produce. In general, however, the high-entropy instances guaranteed by Theorem IV.3 exist non-constructively (via contraposition), and it remains open whether they can be constructed explicitly – see Section VIII.

## V. ALGORITHMIC POSITIVITY

We now formalize the No-Go framework.

### A. Definition

**Definition V.1** (Algorithmic Positivity). An NP language  $L$  (with verifier  $V$ ) is *algorithmically positive* if there exists a witness-producing function  $W : \{0, 1\}^* \rightarrow \{0, 1\}^*$  satisfying (AP1) and (AP2):

**(AP1) Polynomial scaling.**  $W$  is computable in time  $\text{poly}(|x|)$ , and for every  $x \in L$ :  $V(x, W(x)) = 1$ .

**(AP2) Entropic convexity.** There exists a polynomial  $q$  such that for every  $x \in L$ :

$$K^{q(|x|)}(W(x) \mid x) \leq O(\log |x|). \quad (11)$$

The witness production is “entropically cheap”: the witness  $W(x)$  adds at most logarithmic resource-bounded information beyond the input. (Note: since  $W$  is a fixed polynomial-time program, the polynomial  $q$  is determined by  $W$  and does not depend on the instance  $x$ .)

**Remark V.2.** Note that (AP2) targets the witness  $W(x)$ , not just the decision bit. A single decision bit  $D(x) \in \{0, 1\}$  trivially satisfies  $K(D(x) \mid x) \leq 1$ , so entropic convexity would be vacuous for decision functions. The meaningful measure is the resource-bounded complexity of producing a witness – this is where the information-theoretic content of NP-hardness resides.

**Remark V.3.** Every P-decidable NP language is algorithmically positive: if  $L \in P \cap NP$ , then the polynomial-time search algorithm (obtained via self-reducibility for NP-complete problems, or directly for natural NP problems) produces a witness  $W(x)$  with  $K^{q(|x|)}(W(x) \mid x) \leq |W_{\text{prog}}| + O(\log |x|) = O(\log |x|)$ .

**Remark V.4** (Heuristic properties of P-decidable NP languages). For NP languages in P, the witness-producing function  $W$  additionally satisfies two heuristic properties that capture the “structural simplicity” of polynomial-time witness extraction:

**(H1) Compositional stability.** For structured instances under a well-defined composition operator  $\circ$ , the witness for the composed input is typically determined by the component witnesses with at most logarithmic overhead:  $K^{\text{poly}}(W(x_1 \circ x_2) \mid W(x_1), W(x_2)) \leq O(\log(|x_1| + |x_2|))$ .

**(H2) Monotone stability.** Small perturbations of the input produce witnesses of bounded additional complexity:  $d_H(x, x') \leq 1 \implies K(W(x) \mid W(x'), x) \leq O(\text{poly}(\log |x|))$ .

These properties are motivated by the structure of NP-complete problems but are not used in the formal No-Go argument (Theorem VI.1). They serve as additional heuristic evidence for the Entropic Separation Conjecture.

### B. The Entropic Separation Conjecture

**Conjecture V.5** (Entropic Separation – ESC). No NP-complete language is algorithmically positive. That is: for every NP-complete language  $L$  and every polynomial-time witness-producing function  $W$ , there exist infinitely many yes-instances  $x \in L$  such that  $W$  fails to produce a valid witness for  $x$  (i.e.,  $V(x, W(x)) = 0$ ), or  $K^{q(|x|)}(W(x) \mid x) \geq \omega(\log |x|)$  for all polynomials  $q$ .

A stronger parallel truth-table bridge target (see Theorem VII.4) is: for some NP-complete language  $L$ , for every polynomial  $q$ , infinitely many  $n$  satisfy  $K^{q(2^n)}(L_n) \geq n$ . This truth-table statement is not claimed to be equivalent to ESC as stated here; Theorem VII.4 shows that it is a sufficient route to  $P \neq NP$  and a circuit-lower-bound-style bridge target.

**Proposition V.6.** *The Entropic Separation Conjecture V.5 implies  $P \neq NP$ .*

*Proof.* If  $P = NP$ , then for every NP-complete language  $L$ , the search version is solvable in polynomial time (for SAT, by self-reducibility; for general NP-complete  $L$ , by reducing  $L$  to SAT, applying the SAT search algorithm, and reconstructing the witness). This gives a polynomial-time witness-producing function  $W$  satisfying (AP1). Since  $W$  is a fixed program running in polynomial time,  $K^{q(|x|)}(W(x) \mid x) \leq |W_{\text{prog}}| + O(\log |x|) = O(\log |x|)$ , satisfying (AP2). Therefore, NP-complete languages would be algorithmically positive, contradicting the conjecture.  $\square$

*Remark V.7* (ESC is equivalent to  $P \neq NP$ ). The converse of Proposition V.6 also holds:  $P \neq NP$  implies ESC. Equivalently, argue by contraposition: if ESC fails, then some NP-complete language has a polynomial-time witness-producing function satisfying (AP1) and (AP2). This is already a polynomial-time search procedure for an NP-complete language, and the standard search-to-decision reduction yields  $P = NP$ . Thus  $\text{ESC} \Leftrightarrow P \neq NP$ : the Entropic Separation Conjecture is a *reformulation* of the  $P$  vs  $NP$  problem in information-theoretic language, not an independent strengthening. The truth-table version below is a stronger sufficient bridge target using Lemma VII.3.

*Priority note:* This equivalence is a reformulation of the known equivalence between  $P \neq NP$  and unbounded time-bounded instance complexity of SAT, established by Fortnow [24] building on the instance complexity framework of Orponen et al. [25]. Our contribution is the specific formulation in terms of conditional witness entropy  $K^t(w \mid x)$  rather than instance complexity  $ic_t(x : \text{SAT})$ .

### C. Why NP-complete problems resist algorithmic positivity

The failure of each condition for NP-complete problems:

**(AP2) fails:** For NP-complete problems, producing a witness  $w$  for  $x \in L$  requires extracting information that is not uniformly logarithmically compressible relative to the instance. Theorem IV.3 shows, conditional on  $P \neq NP$ , that for every fixed polynomial time bound and logarithmic constant, infinitely many SAT instances have no valid witness within that description budget. The stronger near-maximal bound  $K^t(w \mid x) = \Omega(|w|)$  is an additional bridge hypothesis, not a theorem. No polynomial-time witness-producing function  $W$  can achieve  $K^{\text{poly}}(W(x) \mid x) \leq O(\log |x|)$  for all SAT instances.

As additional heuristic evidence (not part of the formal argument), the properties (H1)–(H2) from Remark V.4 also fail for NP-complete problems:

**(H1) fails:** SAT is not compositionally decomposable in the required sense. The satisfiability of a conjunction  $\varphi_1 \wedge \varphi_2$  is not determined by the satisfiability of

$\varphi_1$  and  $\varphi_2$  individually (they share variables). This non-decomposability is precisely the source of NP-hardness.

**(H2) fails for random  $k$ -SAT near the threshold:** For random  $k$ -SAT instances near the satisfiability threshold, SAT exhibits extreme sensitivity in its search version: flipping a single clause can restructure the entire witness space. Specifically, near the phase transition (clause-to-variable ratio  $\alpha \approx \alpha_c(k)$ ), a single clause addition can fragment the solution cluster into exponentially many components, producing witnesses of unbounded additional complexity [29, 30]. **Scope limitation:** This instability is rigorously established only for random  $k$ -SAT in the near-threshold regime  $\alpha \approx \alpha_c(k)$ . For structured, planted, or under-constrained instances ( $\alpha \ll \alpha_c$ ), the witness space may be well-connected, and monotone stability can hold. The heuristic argument for (H2) failure as a general NP-hardness phenomenon thus rests on the assumption that the near-threshold regime is representative of the combinatorial difficulty of NP-complete problems.

## VI. THE NO-GO THEOREM

**Theorem VI.1** (Entropic No-Go – Reformulation). *The following are equivalent:*

- (i)  $P \neq NP$ .
- (ii) *The Entropic Separation Conjecture V.5 holds.*
- (iii) *No NP-complete language is decidable in polynomial time. That is: for every NP-complete language  $L$  and every polynomial-time algorithm  $A$ , there exist instances  $x$  where  $A(x) \neq L(x)$ .*

*The equivalence (i)  $\Leftrightarrow$  (ii) shows that  $P \neq NP$  is equivalent to an entropy obstruction: polynomial-time algorithms cannot compress NP-witnesses below their intrinsic Kolmogorov complexity.*

*Proof.* (ii)  $\Rightarrow$  (i): Shown in Proposition V.6. Contrapositive: if  $P = NP$ , then every NP-complete language has a polynomial-time witness-producing function (via the SAT search algorithm and polynomial-time reductions), satisfying both (AP1) and (AP2). This makes NP-complete languages algorithmically positive, contradicting ESC.

(i)  $\Rightarrow$  (ii): Contrapositive: suppose ESC fails, i.e., some NP-complete language  $L$  is algorithmically positive. Then  $L$  has a polynomial-time witness-producing function  $W$  with  $K^{\text{poly}}(W(x) \mid x) \leq O(\log |x|)$ . In particular,  $W$  is a polynomial-time search algorithm for  $L$ . Since  $L$  is NP-complete, this implies  $P = NP$ .

(i)  $\Leftrightarrow$  (iii): (iii) states that for every NP-complete language  $L$ ,  $L \notin P$ . This is equivalent to  $P \neq NP$  by the existence of NP-complete languages (if any NP-complete  $L$  were in  $P$ , then  $P = NP$ ; conversely,  $P = NP$  would place every NP-complete language in  $P$ ).

The content of the theorem is not a new separation result, but a *unified presentation* of the known equivalence



between  $P \neq NP$  and entropic incompressibility of NP-witnesses (cf. Fortnow [24], Orponen et al. [25]). The specific framing in terms of algorithmic positivity and the No-Go structure is the contribution of this paper.  $\square$

### A. Exclusion of alternative scenarios

**Proposition VI.2** (Excluded scenarios). *The following scenarios are incompatible with the conjunction of entropic positivity and Kolmogorov obstruction:*

(X1) **Polynomial algorithm with exponential entropy production.** *An algorithm that “invents” new information during computation. Excluded because for any deterministic algorithm  $A$  running in time  $t_A(|x|)$ , the resource-bounded conditional complexity satisfies  $K^{t_A(|x|)}(A(x) \mid x) \leq |A| + O(\log |x|)$ : the program “run  $A$  on input  $x$ ” has constant description length. In particular, the conditional algorithmic information of  $A(x)$  given  $x$  is at most  $O(1)$  bits at the time scale of  $A$ ’s own computation – meaning  $A$  cannot “create” information that was not already implicit in the input. (For randomized algorithms, the random bits  $r$  provide additional input, so  $K(A(x, r) \mid x)$  can be high. However, the  $P$  vs  $NP$  question concerns deterministic computation; moreover, under widely believed derandomization assumptions,  $BPP = P$  [15], so randomness does not provide additional power.)*

(X2) **Compositional collapse of NP.** *All SAT instances decompose into independently solvable parts. Excluded by the NP-completeness of SAT: if SAT decomposed compositionally, then all NP problems would decompose (via reduction). However, general SAT instances involve shared variables across clauses, so the satisfiability of a conjunction  $\varphi_1 \wedge \varphi_2$  is not determined by the satisfiability of the parts individually. The Valiant–Vazirani theorem [21] further strengthens this: SAT reduces (under randomized reductions) to instances with a unique satisfying assignment, where compositional decomposition is impossible since the unique witness encodes global correlations across all variables.*

(X3) **Smooth decision landscape.** *All SAT instances have monotonically stable decision boundaries. Excluded by phase-transition phenomena in random SAT: near the satisfiability threshold, the decision changes discontinuously under small clause additions.*

## VII. RESOURCE-BOUNDED SLICE ENTROPY

The witness entropy gap of Section IV targets individual instances. A cleaner formulation targets the *truth table* of the language on all inputs of a given length.

### A. Slices of a language

**Definition VII.1** (Slice). For a language  $L \subseteq \{0, 1\}^*$ , the  $n$ -th *slice*  $L_n \in \{0, 1\}^{2^n}$  is the truth table of  $L$  on inputs of length  $n$ :  $(L_n)_i = 1 \iff x_i \in L$ , where  $x_1, \dots, x_{2^n}$  are the  $n$ -bit strings in lexicographic order.

**Definition VII.2** (Resource-bounded algorithmic entropy). For a time bound  $t(\cdot)$ :

$$K^t(x) := \min\{|p| : U(p) = x \text{ in at most } t(|x|) \text{ steps}\}. \quad (12)$$

### B. The slice entropy lemma

**Lemma VII.3** ( $P$  implies low slice entropy). *If  $L \in P$ , then there exists a polynomial  $q$  and a constant  $c$  such that for all  $n$ :*

$$K^{q(2^n)}(L_n) \leq c + \lceil \log n \rceil. \quad (13)$$

*Proof.* Let  $A$  be a polynomial-time algorithm deciding  $L$  in time  $n^d$ . The program  $p$ : “read  $n$  from input; for each  $x \in \{0, 1\}^n$ , run  $A(x)$  and output the result bit” has length  $|A| + O(\log n)$  (to encode  $n$ ). Its runtime is  $2^n \cdot n^d = \text{poly}(2^n)$ . Thus  $K^{q(2^n)}(L_n) \leq |A| + O(\log n) = c + \lceil \log n \rceil$ .  $\square$

### C. The No-Go theorem via slice entropy

**Theorem VII.4** (Slice Entropy No-Go). *If there exists an NP-complete language  $L$  satisfying the Entropy Hardness Hypothesis:*

(EH) *For every polynomial  $q$ , there exist infinitely many  $n$  with*

$$K^{q(2^n)}(L_n) \geq n, \quad (14)$$

*then  $P \neq NP$ .*

*Proof.* Immediate from Lemma VII.3:  $L \in P$  would give  $K^{\text{poly}(2^n)}(L_n) = O(\log n)$ , contradicting (14) for large  $n$ .  $\square$

**Remark VII.5.** The Entropy Hardness Hypothesis (EH) is a concrete, well-defined conjecture about a specific NP-complete language (e.g., SAT). It asserts that the truth table of SAT on  $n$ -bit formulas cannot be compressed to fewer than  $n$  bits, even when the compression algorithm is allowed  $\text{poly}(2^n)$  time.

This is a clean sufficient No-Go formulation: the “easy direction” ( $L \in P \Rightarrow$  low entropy) is a theorem; the “hard direction” (EH for SAT) is a stronger truth-table/circuit-lower-bound route to  $P \neq NP$ , not a consequence of  $P \neq NP$  alone.

Note that the threshold  $n$  in EH is much smaller than the truth-table length  $2^n$ : a random string of length  $2^n$  has  $K \approx 2^n$ . The threshold  $n$  is chosen because

Lemma VII.3 gives  $K^{\text{poly}(2^n)}(L_n) = O(\log n)$  for  $L \in \mathbf{P}$ ; any threshold  $\omega(\log n)$  would suffice to separate from  $\mathbf{P}$ , and  $n$  provides a clean, moderate lower bound that is already superpolynomial in the input length.

*Remark VII.6* (Encoding sensitivity of EH). The Entropy Hardness Hypothesis depends on the encoding of SAT instances as binary strings. Different encodings yield truth tables  $\text{SAT}_n$  of different lengths and potentially different Kolmogorov complexities. The hypothesis should be understood as: there exists a *standard* polynomial-time encoding  $\text{enc} : \text{Formulas} \rightarrow \{0, 1\}^*$  such that  $K^{q(2^n)}(\text{SAT}_n^{\text{enc}}) \geq n$  for infinitely many  $n$ . For resource-bounded  $K^t$ , changing the universal machine introduces a polynomial slowdown rather than just an additive constant. However, since EH quantifies over *all* polynomials  $q$ , a polynomial slowdown in the time bound does not affect the hypothesis: if  $K^{q(2^n)}(\text{SAT}_n) \geq n$  for all  $q$ , the same holds after re-parameterization. The *length* of the truth table (and hence the threshold  $n$ ) is encoding-dependent.

*Remark VII.7* (Connection to circuit lower bounds). The Entropy Hardness Hypothesis is related to circuit descriptions, but the threshold  $n$  is too weak to imply standard non-uniform lower bounds by itself. A Boolean function  $f_n$  computed by a circuit of size  $s$  has truth table describable in  $O(s \log s)$  bits, so  $K^{O(s \cdot 2^n)}(f_n) \leq O(s \log s) + O(\log n)$ . Thus:

- EH with threshold  $n$  only rules out circuit descriptions shorter than  $n$  bits. Quantitatively, for circuits used as direct descriptions of  $\text{SAT}_n$ , it implies  $s \log s = \Omega(n)$ , hence only  $s = \Omega(n/\log n)$  up to constants.
- EH does *not* imply  $\text{NP} \not\subseteq \text{P/poly}$  or  $\text{NP} \not\subseteq \text{SIZE}(2^{o(n)})$ . Such conclusions would require much stronger truth-table incompressibility, or an additional theorem converting uniform SAT-slice compression into non-uniform circuit lower bounds.

This keeps the entropy framework aligned with circuit complexity without overstating what the moderate EH threshold proves.

*Remark VII.8* (SAT slice entropy at the phase transition – exploratory numerics). As a purely illustrative exercise (not constituting evidence for or against the ESC), we computed the Shannon entropy  $H_{\text{slice}}$  of the truth table of random 3-SAT at the phase transition  $\alpha \approx 4.267$  for  $n = 8, 10, 12, 14, 16$ . The entropy grows from  $H \approx 1.5$  ( $n = 8$ ) to  $H \approx 2.3$  ( $n = 14$ ). **Limitations:** (i) Only 5 data points are available, which is far too few to determine asymptotic growth behavior – any monotone function (including logarithmic growth, which would *refute* the ESC) is consistent with these data. (ii) The Shannon entropy  $H_{\text{slice}}$  of the truth table is *not* the same as the Kolmogorov complexity  $K^t(\text{SAT}_n)$ ; the relationship is not formally established here. (iii)  $n \leq 16$  is far below the asymptotic regime relevant to the ESC. Full validation

would require  $n \geq 30$ , which is computationally infeasible. Script: `compute_sat_entropy.py`.

#### D. Why (EH) should be true

The truth table  $\text{SAT}_n$  has length  $2^n$  and describes the satisfiability of all  $n$ -bit encodings of Boolean formulas. There are  $2^{2^n}$  possible truth tables of this length, and  $\text{SAT}_n$  is a *specific* one determined by a complete mathematical definition (SAT). Nevertheless:

1. **Pseudorandom structure.** The satisfiability boundary in the space of formulas exhibits phase-transition behavior [14]: near the threshold, the pattern of satisfiable vs. unsatisfiable instances has high apparent randomness.
2. **No known compression.** Despite decades of SAT-solver research, no method compresses  $\text{SAT}_n$  to  $o(2^n)$  bits in  $\text{poly}(2^n)$  time. The best algorithms (CDCL solvers) run in time  $2^{\Theta(n)}$  in the worst case.
3. **Hardness-vs-randomness evidence.** Explicit truth-table incompressibility is closely tied to circuit lower bounds and derandomization [15]. Thus EH is a natural strengthening to test in the same ecosystem as PRG and MCSP routes, but its failure would only close this particular truth-table route, not refute  $\mathbf{P} \neq \mathbf{NP}$ .

### VIII. THE PRECISE OBSTRUCTION

#### A. The uniformity gap

The main argument (Section IV) has a precise gap that we now identify.

**What is proven:** There *exist* NP instances whose witnesses have high conditional Kolmogorov complexity (Theorem IV.3). This is a *non-uniform* statement about individual strings.

**What is needed:** A *uniform* statement: for every polynomial-time algorithm  $A$ , there exist infinitely many instances  $x$  where  $A(x)$  fails. This requires converting the existential entropy bound into an *adversarial* bound against all efficient algorithms simultaneously.

**Definition VIII.1** (The uniformity bridge). Let  $L$  be an NP-complete language with verifier  $V$ . The *uniformity bridge* is the statement: for every polynomial-time computable function  $A : \{0, 1\}^* \rightarrow \{0, 1\}^*$ , there exist infinitely many yes-instances  $x \in L$  such that  $A$  fails to produce a valid witness:

$$\forall A \in \text{FP}, \exists^\infty x \in L : V(x, A(x)) = 0. \quad (15)$$

Equivalently, in entropic terms: for every polynomial-time  $A$  and every polynomial  $q$ , there exist infinitely

many  $x \in L$  such that every valid witness  $w$  for  $x$  satisfies  $K^{q(|x|)}(w \mid x) > |A| + O(\log |x|)$ .

Note: the uniformity bridge is formulated as a search statement ( $A$  must produce a *witness*, i.e.,  $A \in \text{FP}$  and  $V(x, A(x)) = 1$ ). The connection to the decision problem  $P \neq \text{NP}$  follows from the NP-completeness of  $L$ : for NP-complete languages, the search and decision versions are polynomial-time equivalent (via self-reducibility for SAT, or more generally via Cook–Levin plus polynomial-time reductions).

The uniformity bridge is the P-vs-NP analog of:

- The “higher Gross–Zagier formula” in BSD (coupling analytic rank to algebraic rank).
- The “hard direction” in the Hodge conjecture (showing positivity of the Hodge–Riemann form implies algebraicity).

In each case, the positivity structure is established, but a *coupling theorem* is needed to convert positivity into a definitive result.

## B. Known partial results

The uniformity bridge is partially established in restricted settings:

1. **Resolution proof systems:** Exponential lower bounds on resolution refutations of random  $k$ -SAT instances near the threshold [6]. This proves that resolution-based SAT solvers cannot solve these instances efficiently.
2. **Bounded-depth circuits:**  $\text{SAT} \notin \text{AC}^0$  (Furst–Saxe–Sipser 1984, Ajtai 1983; see [19] for optimal bounds). This proves the entropy obstruction for constant-depth circuits.
3. **Monotone circuits:** Exponential lower bounds for monotone circuits computing CLIQUE [7]; related bounded-depth lower bounds [8].
4. **Algebraic proof systems:** Exponential lower bounds for Nullstellensatz and Polynomial Calculus refutations [9].

Each of these is a *model-specific* instantiation of the entropy obstruction. The challenge is to prove it for *general* polynomial-time computation – which is precisely the  $P \neq \text{NP}$  problem.

## C. Why the gap is hard

The uniformity gap is hard for the same reason that P vs NP is hard: it requires a statement about *all* polynomial-time algorithms. In the uniform model, this is a universal quantifier over a countably infinite, recursively enumerable

set (all Turing machines running in polynomial time). In the non-uniform model (P/poly), the quantifier ranges over all polynomial-size circuit families, an uncountable set.

The entropy approach *reformulates* the problem but does not *simplify* it:

*P vs NP is equivalent to asking whether Kolmogorov complexity can be polynomially compressed for NP-witness structures [24]. The entropy framework makes this equivalence explicit in the language of witness entropy and identifies concrete bridge strategies, but converting this to a proof against all algorithms remains the central challenge – which is P vs NP itself in different language.*

## D. Four candidate bridge strategies

We identify four concrete research directions for closing the uniformity gap (summarized with their core mechanisms and technical bottlenecks in Table IV):

**(B1) Dense families with unique random witnesses.** Construct *dense* families  $S_n \subseteq \{0, 1\}^{\text{poly}(n)}$  with  $|S_n|/2^{\text{poly}(n)}$  non-vanishing, such that every  $x \in S_n$  has a unique valid witness  $w_x$  with  $K^{\text{poly}(n)}(w_x \mid x) \geq |w_x| - O(\log n)$ . Density ensures that no polynomial-time algorithm can “avoid” all hard instances. The construction requires efficient reductions from Kolmogorov-random strings to SAT instances with unique solutions – a non-trivial but plausible coding-theoretic task.

**(B2) Self-referential diagonalization.** For each polynomial-time machine  $A$ , construct an instance  $x_A$  whose unique valid witness  $w$  contains (or encodes) a description of  $A$  itself, so that  $K^{\text{poly}(|x_A|)}(w \mid x_A) \geq |A|$ . This creates a self-referential obstruction:  $A$  cannot produce a witness that is “about itself” without exceeding its own description length. The approach combines Gödel-style diagonalization with Kolmogorov complexity and avoids relativization because the self-reference targets the machine’s *description*, not its oracle behavior.

**(B3) Resource-bounded slice complexity.** Prove that for SAT (or another NP-complete language), the resource-bounded complexity satisfies  $K^{\text{poly}(2^n)}(\text{SAT}_n) \geq 2^{n-o(n)}$  for infinitely many  $n$ . This is a *uniform* statement about the truth table and directly implies  $P \neq \text{NP}$  via the Slice Entropy No-Go (Theorem VII.4). The strategy connects to proof complexity: if every short proof of an instance’s satisfiability has high Kolmogorov complexity, then no polynomial-time algorithm can systematically produce such proofs.

**(B4) Minimum Circuit Size Problem (MCSP).** Recent work connects resource-bounded Kolmogorov complexity to circuit complexity via the *Minimum Circuit Size Problem* (MCSP): given a truth table  $t \in \{0, 1\}^{2^n}$  and a parameter  $s$ , decide whether there exists a circuit of size  $\leq s$  computing  $t$ . Allender et al. [18] established deep

Table IV. Taxonomy of candidate strategies for bridging the uniformity gap.

| Strategy                    | Core Mathematical Mechanism                                                    | Key Technical Bottleneck                                                                 | Barrier Status                                    | Paper Connection                    |
|-----------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------|
| (B1) Dense Families         | Reductions from random strings to SAT instances                                | Ensuring non-vanishing density $ S_n /2^{\text{poly}(n)}$                                | Non-relativizing by construction                  | Theorem IV.3 (Witness Entropy Gap)  |
| (B2) Self-Referential Diag. | Encoding machine description $ A $ into witness $w$                            | Superpolynomial length gap maintenance                                                   | Targets description, not oracle                   | Diagonalization avoiding BGS        |
| (B3) Slice Complexity       | Uniform truth-table bound $K^{\text{poly}(2^n)}(\text{SAT}_n) \geq 2^{n-o(n)}$ | Non-constructive lower bounds on slices                                                  | Bypasses natural proofs via slice incomputability | Theorem VII.4 (Slice Entropy No-Go) |
| (B4) MCSP Hardness          | Circuit minimization hardness via $K^t$ connections                            | Quantitative losses along $\text{MCSP} \Rightarrow \text{circuit} \Rightarrow \text{EH}$ | Connects $K^t$ to derandomization                 | Allender et al. [18], Rem. XI.20    |

connections between MCSP and  $K^t$ . While  $\text{MCSP} \in \text{NP}$ , its complexity status is open: it is not known to be NP-complete, yet proving  $\text{MCSP} \in \text{P}$  would have strong derandomization consequences. Strong MCSP lower bounds or hardness results may feed into circuit-lower-bound programs; they imply the Entropy Hardness Hypothesis only after an additional step that yields sufficiently strong SAT-slice truth-table incompressibility. The chain is therefore conditional rather than automatic:  $\text{MCSP hardness} \Rightarrow \text{circuit lower bounds} \Rightarrow \text{truth-table incompressibility} \Rightarrow \text{EH}$ , with quantitative losses at each arrow.

## IX. PROOF COMPLEXITY AS A TESTING GROUND

### A. SAT algorithms and short proofs

A deep connection exists between algorithms and proofs:

$$\text{SAT} \in \text{P} \implies \begin{array}{l} \text{every unsatisfiable formula} \\ \text{has a short refutation,} \end{array} \quad (16)$$

in a proof system at least as strong as Extended Frege.

**Proposition IX.1** (Proof complexity formulation – Cook [5]). *If  $\text{P} \neq \text{NP}$ , then there is no polynomially bounded proof system for tautologies. Equivalently:  $\text{P} = \text{NP}$  if and only if there exists a proof system in which every tautology of length  $n$  has a proof of length  $\text{poly}(n)$ .*

This is Cook’s reformulation [5]: proving superpolynomial lower bounds on proof length in every propositional proof system is equivalent to showing  $\text{P} \neq \text{coNP}$  (which follows from  $\text{P} \neq \text{NP}$ ).

### B. Entropy of proofs

The entropy framework connects to proof complexity via the following question: for a random unsatisfiable formula  $\varphi$  and its shortest refutation  $\pi$  in a given proof system, what is  $K(\pi \mid \varphi)$ ? If  $\pi$  is “algorithmically random” relative to  $\varphi$  – meaning  $K(\pi \mid \varphi)$  is close to  $|\pi|$  – then no efficient algorithm can systematically produce such refutations.

This provides a concrete testing ground: if one can show that for random unsatisfiable 3-SAT near the threshold, the shortest refutation in Extended Frege has both superpolynomial length and high conditional Kolmogorov complexity, this would be strong evidence for the entropy obstruction.

**Conjecture IX.2** (Resolution width and Kolmogorov complexity). *For unsatisfiable CNF formulas  $\varphi_n$  on  $n$  variables, the minimum resolution width  $w(\varphi_n \vdash \perp)$  satisfies*

$$w(\varphi_n \vdash \perp) \geq \Omega(K(\pi^* \mid \varphi_n) / \log n),$$

where  $\pi^*$  is the shortest resolution refutation of  $\varphi_n$ . In particular, formulas whose refutations have high Kolmogorov complexity require wide clauses.

*Motivation.* Ben-Sasson and Wigderson [6] showed that resolution width lower bounds imply resolution length lower bounds via  $|\pi| \geq 2^{w-O(n)}$ . Our conjecture reverses the direction: high complexity of the proof object (measured by  $K$ ) forces width. If true, this would provide an information-theoretic characterization of resolution width, connecting proof complexity to the ESC framework.

*Remark IX.3* (Resolution width as thermodynamic observable). Resolution width  $w(\varphi_n \vdash \perp)$  is robust under time-bounding: unlike Kolmogorov complexity  $K$ , the width of a Resolution refutation is a *syntactic* invariant that does not depend on computational resources. This makes it a natural “thermodynamic observable” in the proof-complexity analogy: large width forces any time-bounded description to be complex, providing a concrete route from  $K$ -lower bounds to  $K^t$ -lower bounds for specific formula families. The conditional MCSP hardness results of Hirahara and Ilango [37] motivate MCSP as a bridge object, without requiring general decidability of  $K$ .

### C. Entropy proof of $\text{SAT} \notin \text{AC}^0$

As a concrete instantiation of the entropy framework, we give an entropy-based argument for the classical result  $\text{PAR} \notin \text{AC}^0$  (Furst–Saxe–Sipser 1984, Ajtai 1983). Since  $\text{AC}^0$  is closed under  $\text{AC}^0$ -reductions and parity is  $\text{AC}^0$ -reducible to SAT (via a standard reduction that encodes the parity constraint as a CNF formula using constant-depth circuitry), this also implies  $\text{SAT} \notin \text{AC}^0$ .

**Proposition IX.4** (Entropy obstruction for  $\text{AC}^0$ ). *Let  $\{C_n\}$  be a constant-depth circuit family of polynomial size  $s(n) = n^{O(1)}$  computing a Boolean function  $f_n : \{0, 1\}^n \rightarrow \{0, 1\}$ . Then the truth table  $f_n \in \{0, 1\}^{2^n}$  satisfies*

$$K^{s(n) \cdot 2^n}(f_n) \leq O(n^c \log n) \quad (17)$$

for some constant  $c$  depending on the depth and size bound. In contrast, a random Boolean function  $g_n$  satisfies  $K(g_n) \geq 2^n - O(1)$  with high probability. The entropy gap between  $\text{AC}^0$ -computable functions and random functions is thus exponential:  $O(n^c \log n)$  vs.  $2^n$ .

*Proof sketch.* A depth- $d$  circuit of size  $s$  on  $n$  inputs is described by  $O(s \cdot \log s)$  bits (gate types and wiring). Given this description, the truth table on all  $2^n$  inputs is computed in time  $O(s \cdot 2^n)$  by brute-force evaluation. Thus  $K^{O(s \cdot 2^n)}(f_n) \leq O(s \cdot \log s) + O(\log n)$ . For polynomial size  $s = n^c$ :  $K^{n^c \cdot 2^n}(f_n) \leq O(n^c \log n)$ .

The classical result  $\text{PAR}_n \notin \text{AC}^0$  (Furst–Saxe–Sipser 1984, Ajtai 1983, Håstad 1987) shows that the parity function cannot be computed by polynomial-size constant-depth circuits. In the entropy framework, this means: any circuit computing  $\text{PAR}_n$  requires super-polynomial size  $s(n) = 2^{n^{\Omega(1/d)}}$ , so  $K^{\text{poly}(2^n)}(\text{PAR}_n)$  is *not* bounded by  $O(n^c \log n)$  when computed via  $\text{AC}^0$  circuits. The entropy method does not *reprove*  $\text{PAR} \notin \text{AC}^0$ , but it *reframes* the separation:  $\text{AC}^0$ -computable truth tables occupy a negligible fraction of the space of all Boolean functions, and parity lies outside this low-entropy region.  $\square$

This demonstrates the entropy perspective on a known separation. For general P/poly, the analogous bound would give  $K^{\text{poly}(2^n)}(f_n) \leq O(n^c \log n)$  for any  $f \in \text{P/poly}$ . Proving that  $\text{SAT}_n$  exceeds this bound would establish  $\text{NP} \not\subseteq \text{P/poly}$  – a major open problem that implies  $\text{P} \neq \text{NP}$ .

*Remark IX.5* (Håstad’s bound as entropy capacity). Håstad’s switching lemma [19] can be interpreted as an *information-theoretic capacity bound*: after applying a random restriction that fixes each variable independently with probability  $1 - p$ , a depth- $d$   $\text{AC}^0$  circuit on the surviving variables simplifies to a decision tree of depth  $O(\log n)^{d-1}$ . In entropy terms: each layer of an  $\text{AC}^0$  circuit can “aggregate” at most  $O(\log n)$  bits of information from its inputs (via the fan-in bound). After  $d$  layers, the total information capacity is  $O((\log n)^d)$  bits – far below the  $\Omega(n)$  bits needed to compute parity. This capacity bound  $O((\log n)^d) \ll n$  is the entropy-theoretic core of the  $\text{PAR} \notin \text{AC}^0$  separation.

## D. Geometric Complexity Theory

Mulmuley’s Geometric Complexity Theory (GCT) [10] approaches P vs NP through algebraic geometry and representation theory. The GCT program seeks *obstructions* – representation-theoretic invariants that distinguish easy from hard functions.

In the entropy framework, GCT obstructions are a specific type of *algebraic entropy certificate*: they certify that a function has high “algebraic complexity” by exhibiting an invariant that polynomial-size circuits cannot match.

The connection to algorithmic positivity: GCT obstructions are *non-natural* (they are not efficiently computable properties of truth tables) and *non-relativizing* (they depend on algebraic structure). They thus satisfy specifications (S1)–(S2). Whether they satisfy (S3) (non-algebrization) is an active research question.

## X. RELATED WORK

The connection between Kolmogorov complexity and computational complexity has been studied extensively since the 1990s. The equivalence between time-bounded Kolmogorov complexity and the P vs NP question is *not* new to this paper; our contribution lies in the specific framing as a No-Go theorem via algorithmic positivity, the systematic barrier analysis, and the identification of concrete bridge strategies. We briefly situate our work within this literature.

**Instance complexity and the P vs NP equivalence.** Orponen, Ko, Schöning, and Watanabe [25] introduced instance complexity as a measure for the computational difficulty of individual instances: the  $t$ -bounded instance complexity  $\text{ic}_t(x : A)$  is the length of the shortest program that runs in time  $t$ , decides  $x$  correctly with respect to  $A$ , and makes no mistakes on other inputs. Fortnow [24] showed explicitly that  $\text{P} \neq \text{NP}$  is equivalent to the statement that the time-bounded instance complexity of SAT is unbounded. The Entropic Separation Conjecture (ESC) of this paper is a reformulation of this known equivalence in the language of conditional witness entropy rather than instance complexity. Our Theorem IV.3 and Proposition V.6 are thus new *formulations* of known results, not new results in themselves.

**Resource-bounded Kolmogorov complexity and circuit lower bounds.** Allender et al. [18] established deep connections between the Minimum Circuit Size Problem (MCSP), resource-bounded Kolmogorov complexity  $K^t$ , and derandomization. They showed that  $K^t$  can encode significant information about circuit complexity: a truth table  $f_n$  computed by circuits of size  $s$  satisfies  $K^{O(s \cdot 2^n)}(f_n) \leq O(s \log s)$  (the time bound  $O(s \cdot 2^n)$  accounts for evaluating the circuit on all  $2^n$  inputs), linking slice entropy directly to circuit lower bounds. Allender [23] surveyed how Kolmogorov complexity and derandomization interact, arguing that understanding the complexity of “random strings” (strings with high  $K^t$ ) is central to resolving P vs NP.

**Uniform lower bounds via probabilistic  $K^t$ .** Santhanam [26] proposed an algorithmic approach to uniform lower bounds using probabilistic time-bounded Kolmogorov complexity ( $\text{pK}^{\text{poly}}$ ). His approach yields *actual new results* (uniform  $\text{AC}^0$  lower bounds) from non-trivial sampling algorithms, without relying on hierarchy theorems. The uniformity bridge of our paper (Section XIV) addresses the same conceptual gap that Santhanam tackles with concrete algorithmic tools; his work demonstrates that the gap is partially closable in restricted settings.

**MCSP and NP-hardness.** Hirahara [27] showed that the NP-hardness of MCSP under certain reductions would imply new circuit lower bounds, providing a concrete path from  $K^t$ -based hardness to P vs NP separations. The chain “hardness of MCSP  $\Rightarrow$  circuit lower bounds  $\Rightarrow$  truth-table incompressibility  $\Rightarrow$  EH” motivates our bridge strategy B4.

**Natural proofs and pseudorandomness.** The nat-

ural proofs barrier [2] is closely connected to the non-computability of  $K$ : a “natural” property must be efficiently computable, but high Kolmogorov complexity is not. Our approach exploits this connection systematically. Impagliazzo and Wigderson [15] showed that strong circuit lower bounds imply derandomization ( $P = BPP$ ), which in Impagliazzo’s “Five Worlds” framework [28] corresponds to Heuristica or beyond. EH should therefore be read as a strong truth-table bridge target in the hardness-vs-randomness ecosystem; failure of EH would not by itself place us in Algorithmica.

**One-way functions and  $K^t$ .** Liu and Pass [45] proved the landmark result that one-way functions exist if and only if  $K^t$  is average-case hard (for  $t(n) \geq (1 + \varepsilon)n$ ). This is the precise connection between resource-bounded Kolmogorov complexity and cryptographic assumptions that underlies our Bridge Target 3 and the MCSP internalization route (Remark XI.20).

**Five Worlds and ESC.** The ESC can be situated within Impagliazzo’s “Five Worlds” framework [28]: ESC holds in Heuristica, Pessiland, Minicrypt, and Cryptomania, and is false only in Algorithmica (where  $P = NP$ ). The min-entropy variant (Remark XI.17) distinguishes further: in Pessiland, average-case NP-hardness holds but OWFs do not exist, so the distributional ESC holds but the PRG-based Bridge Target 3 is unavailable. This motivates Bridge Targets 1 and 2 as alternatives that do not require cryptographic assumptions.

**Distinction from prior work.** This paper does not claim to establish new connections between Kolmogorov complexity and  $P$  vs  $NP$  – such connections have been known since the work of Orponen et al. [25] and Fortnow [24], and have been developed further by Allender [18, 23], Hirahara [27], Liu–Pass [45], and Santhanam [26]. Our contribution is the specific *presentation*: the framing as a No-Go theorem via algorithmic positivity, the systematic barrier analysis (rigorous for  $K$ , heuristic for  $K^t$ ), the identification of the uniformity bridge as the precise remaining gap, and the connection to proof complexity via the resolution-width conjecture.

## XI. DISCUSSION

### A. What has been achieved

1. **Unified presentation.** The known equivalence between  $P \neq NP$  and resource-bounded incompressibility of NP-witnesses [24, 25] is presented in the unified language of algorithmic positivity and entropic No-Go theorems.
2. **Barrier analysis.** Arguments based on *unbounded* Kolmogorov complexity  $K$  satisfy all five barrier specifications: non-relativizing, non-natural, non-algebrizing, model-independent, and robust. For the *resource-bounded* variant  $K^t$  used in the core results, the barrier status is heuristic:  $K^t$  is computable, so

it does not automatically escape the natural proofs barrier, and formal non-algebrization remains open (see Remark III.6).

3. **Witness entropy gap.** Conditional on  $P \neq NP$ , there exist SAT instances whose valid witnesses are not logarithmically compressible within any fixed polynomial time bound and logarithmic constant. The stronger claim that witnesses have near-maximal conditional Kolmogorov complexity is isolated as Conjecture IV.4.
4. **Algorithmic positivity.** The concept identifies two core structural properties (polynomial-time witness extraction and entropic cheapness of witnesses) that  $P$ -decidable NP languages satisfy and NP-complete problems violate, supplemented by heuristic properties (compositional and monotone stability of witnesses) that provide additional separation evidence.

### B. What remains open

The central open problem – the *uniformity bridge* – is the conversion of existential entropy bounds (“there exist hard instances”) to universal algorithm bounds (“no algorithm solves all instances”). At a high level, this gap – between establishing a non-constructive property and closing the loop with a constructive coupling – appears in other mathematical contexts as well (see Table V), though the analogy is at the level of proof strategy, not mathematical content:

Table V. Structural analogy of non-negative quantities and missing couplings across mathematical and complexity problems.

| Problem | Non-negative quantity | Missing coupling           |
|---------|-----------------------|----------------------------|
| Hodge   | HR-form $\geq 0$      | AP $\Rightarrow$ algebraic |
| BSD     | $\hat{h} \geq 0$      | Higher Gross–Zagier        |
| P vs NP | $K^t(w   x)$ high     | Uniformity bridge          |

### C. Limitations and intellectual honesty

**This paper does not prove  $P \neq NP$ .** We state this unambiguously. The contributions are: (1) a clean reformulation that identifies *what* must be shown (witness incompressibility), (2) a barrier analysis arguing *why* this approach is not blocked by known meta-theorems, and (3) concrete research directions (bridge strategies B1–B4) for closing the remaining gap.

1. **Non-computability of  $K$ .** Kolmogorov complexity is not computable, which means the entropy obstruction cannot be directly “run” as an algorithm. This is a feature (it bypasses natural proofs)

but also a limitation (it makes the argument non-constructive).

2. **The unique-witness requirement.** The entropy gap argument is strongest for instances with unique witnesses. For instances with many witnesses, a polynomial-time algorithm might select a low-complexity witness even if high-complexity witnesses exist.
3. **Equivalence, not implication.** The Entropic Separation Conjecture is logically equivalent to  $P \neq NP$  (Remark V.7). Therefore, this paper’s No-Go theorem is a *reformulation*, not an independent proof strategy. The value lies in identifying the information-theoretic structure of the problem.
4. **Barrier analysis: rigorous for  $K$ , heuristic for  $K^t$ .** For *unbounded*  $K$ , the barrier bypass is well-motivated:  $K$  is non-computable (defeating natural proofs), oracle-independent (defeating relativization), and non-algebraic (plausibly defeating algebraization). However, the core results use *resource-bounded*  $K^t$ , which is computable and whose invariance is only up to polynomial slowdown. For  $K^t$ , the natural proofs barrier is not automatically bypassed, and formal non-algebraization remains open (Remark III.6). Establishing full barrier immunity for  $K^t$ -based arguments is an important research direction.

#### D. The Uniformity Bridge: lemma targets

The central open problem of this paper is the *Uniformity Bridge*: transferring the entropy separation from unbounded Kolmogorov complexity  $K$  (where barrier immunity holds) to resource-bounded complexity  $K^t$  (where computational relevance lies). We identify three concrete lemma targets. In a sufficiently strong and uniform form, any one of them would amount to a conditional closure of the bridge; as stated here they are targets and diagnostics, not established reductions.

*Bridge Target 1: instance compression. Statement:* If no polynomial-time algorithm can compress satisfying assignments of  $\varphi_n$  below  $|\varphi_n|^{1-\varepsilon}$  bits, then  $K^{n^c}(w \mid \varphi_n) \geq |\varphi_n|^{1-\varepsilon}$ , or in the strongest unique-witness form  $K^{n^c}(w \mid \varphi_n) \geq |w| - O(\log n)$ , for every valid witness  $w$ .

*Why it suffices:* Instance compression hardness is a well-studied notion in parameterized complexity [31, 32]. The missing step is a genuine witness-to-compression transfer lemma: a short program that computes  $w$  from  $\varphi_n$  must be converted into a short, verifier-independent representation in the precise sense used by kernelization or OR-compression lower bounds. Such a transfer is not automatic from the mere existence of a witness selector; it is the content of this bridge target.

*Bridge Target 2: proof complexity. Statement:* For any propositional proof system  $P$  and tautology  $\tau_n$  encoding the unsatisfiability of  $\bar{\varphi}_n$ , the proof  $\pi$  satisfies  $K(\pi \mid \tau_n) \geq |\pi|^{1-\varepsilon}$ .

*Why it suffices:* High proof complexity (superpolynomial proof length) implies that proofs themselves carry high entropy. Combined with the Entropic Separation Conjecture, this yields: no polynomial-time proof search can uniformly find proofs, because the proof objects are incompressible. This route connects to the Razborov–Rudich natural proofs barrier in a constructive way.

*Bridge Target 3: PRG-based families. Statement:* There exists an explicit family  $\{\varphi_n\}$  of SAT instances such that (i) each  $\varphi_n$  has a unique satisfying assignment  $w_n$ , (ii)  $w_n$  is the output of a pseudorandom generator  $G : \{0, 1\}^{n/2} \rightarrow \{0, 1\}^n$ , and (iii)  $K^{n^c}(w_n \mid \varphi_n) \geq n/2 - O(\log n)$ .

*Why it suffices:* If  $G$  is a secure PRG (e.g., based on factoring or lattice assumptions), then (iii) follows from the pseudorandomness: any efficient compression of  $w_n$  would break  $G$ . The unique-witness property (i) ensures the entropy gap applies (Corollary IV.5). The equivalence of OWF existence and average-case hardness of  $K^t$  [45] provides a direct cryptographic interpretation: Bridge Target 3 reduces the Uniformity Bridge to a standard cryptographic assumption.

*Remark XI.1 (Minimal sufficient bridge).* Any *one* of the three targets suffices only if it is proved in the strong form stated above. All three targets are themselves major open problems: Target 3 reduces to the existence of secure PRGs (conditional on standard cryptographic assumptions, which are unproven). Target 1 needs a new witness-to-compression transfer lemma before existing instance-compression lower bounds [33] can be applied. Target 2 would require superpolynomial Extended Frege lower bounds, which are widely open. None of these targets should be considered “accessible” in the sense of being within current proof technology; they represent *concrete formulations* of the uniformity gap rather than tractable problems.

*Remark XI.2 (Instance compression as the primary bridge strategy).* We elaborate on Bridge Target 1 as the most promising route, following the “no sparsification unless PH collapses” program of Fortnow–Santhanam [31] and Dell–van Melkebeek [32].

The argument proceeds in three steps:

**Step A: Witness-finder  $\Rightarrow$  low conditional description.** If a polynomial-time algorithm  $A$  produces a valid witness  $w = A(x)$  for  $x \in \text{SAT}$ , then the selected witness has low conditional description relative to  $x$ :  $K^{q_A}(w \mid x) \leq |A| + O(\log |x|)$ . This is the already-proved easy direction of the witness-entropy characterization. It is *not yet* an instance compression of  $x$  in the Fortnow–Santhanam or Dell–van Melkebeek sense. To obtain such a compression one would need an additional map that turns the solver trace, witness information, or verifier certificate into a shorter instance or kernel whose

decompressor is independent of the original instance except through the encoded object.

**Step B: Compression no-go  $\Rightarrow$  uniformity.** The known instance compression impossibility results [31–33] show: if SAT instances of length  $n$  can be compressed to  $n^{1-\varepsilon}$  bits in the appropriate kernelization/OR-compression sense, then the polynomial hierarchy collapses ( $\text{NP} \subseteq \text{coNP/poly}$ ). These results can be invoked here only after Step A has been strengthened to a genuine compression-transfer lemma. Without that strengthening, a witness selector is merely a solver, and using the compression lower-bound theorem would conflate decision compression with witness generation.

**Step C: Translation to  $K^t$ .** If such a compression-transfer lemma were proved and the corresponding compression were impossible (under PH non-collapse), then for infinitely many instances  $x$  every polynomial-time witness production would fail in the logarithmic-description regime. An additional quantitative argument would still be needed to upgrade this to  $K^{n^c}(w \mid x) \geq |w| - O(\log n)$  for all witnesses  $w$ . This quantitative upgrade is the near-maximal bridge, not a consequence of the basic reformulation.

The remaining gaps are therefore twofold: first the witness-to-compression transfer itself, and then the quantifier order. Even a successful Step B would typically give “for each  $A$ , there exist hard instances” (non-uniform), while the Uniformity Bridge needs “there exist instances hard for all  $A$ ” (uniform). Closing this gap requires either (i) a *universal* compression impossibility (not just for each compressor individually), or (ii) a diagonalization argument that uniformizes over all polynomial-time algorithms simultaneously.

*Remark XI.3* (The Uniformity Bridge as a mixing obstruction). The Uniformity Bridge admits a reformulation in the language of statistical mechanics, connecting directly to the Yang–Mills companion paper. Consider the Gibbs measure over witnesses conditioned on an instance  $x$ :

$$\pi_x(w) \propto \mathbf{1}\{V(x, w) = 1\} \cdot e^{-\beta H_x(w)},$$

where  $H_x$  is any energy function on the witness space (e.g.,  $H_x \equiv 0$  for the uniform measure over valid witnesses). Restricted local, random-walk, flip, or DPLL-state algorithms can sometimes be modeled as a *local Markov dynamics* (Glauber/heat-bath type) on a witness-shadow space. A general polynomial-time algorithm need not have such a representation; making it visible as one would itself be a form of the Uniformity Bridge.

For these restricted dynamics one can state the following diagnostic target:

*For every NP-complete language  $L$  and every local Markov dynamics on witnesses, there exist instance families  $\{x_n\}$  for which the mixing time to the witness support is  $\geq \exp(\Omega(n))$ .*

This is a **Dobrushin-type obstruction**: the influence matrix  $C = (c_{ij})$  of the witness-space Gibbs measure has

spectral radius  $\rho(C) \geq 1$  for hard instances, preventing Dobrushin uniqueness and hence rapid mixing. A *superficial* structural parallel exists with the Yang–Mills companion paper [47]: there,  $\rho(C) < 1$  holds for  $\beta < \beta_0$  (strong coupling), yielding a volume-independent mass gap; here,  $\rho(C) \geq 1$  is conjectured for NP-hard instance families. However, this parallel is at the level of the Dobrushin criterion formalism, not at the level of physical or mathematical content – the two settings involve fundamentally different structures.

*Remark XI.4* (Algorithmic Downhill Condition). By analogy with the Downhill Flux Condition (DFC) in the turbulence companion paper [46], define the *Algorithmic Downhill Condition* (ADC): a search process generates a trajectory  $w^{(0)}, w^{(1)}, \dots, w^{(T)}$  of partial witnesses, and a KL-based functional

$$F_x(w^{(k)}) = D_{\text{KL}}(\delta_{w^{(k)}} \parallel \pi_x)$$

measures the “free-energy distance” to the witness distribution. ADC asserts that  $F_x$  is monotonically non-increasing along any polynomial-time trajectory. For NP-complete instances, ADC must be *violated*: the search process requires “uphill steps” (local entropy increase) analogous to backscatter in turbulent cascades. The number of required uphill steps provides a lower bound on the computational complexity of the search—connecting entropic separation to the DFC framework.

*Remark XI.5* (Connection to the Overlap Gap Property). The mixing obstruction of Remark XI.3 has a precise counterpart in the combinatorial optimization literature: the *Overlap Gap Property* (OGP), introduced by Gamarnik and Sudan [38] and developed into a systematic theory of algorithmic barriers [39].

The OGP asserts that for random constraint satisfaction problems near the satisfiability threshold, the solution space exhibits *topological disconnectivity*: the overlap (normalized Hamming distance) between any two near-optimal solutions is either very small or very large, with no intermediate values. This is a geometric analog of the Dobrushin obstruction  $\rho(C) \geq 1$  from Remark XI.3: the Gibbs measure over witnesses may fragment into exponentially many well-separated clusters (the *shattering transition* of Achlioptas and Coja-Oghlan [40]), obstructing broad classes of stable or local dynamics.

The chain of implications is:

$$\begin{aligned} \text{Shattering} &\Rightarrow \text{OGP} \\ &\Rightarrow \text{failure of stable algorithms} \\ &\stackrel{?}{\Rightarrow} \text{high } K^t(w \mid x). \end{aligned}$$

The first two arrows are established only for restricted algorithmic models: Gamarnik, Jagannath, and Wein [41] prove OGP barriers for low-degree polynomials, low-depth/stable Boolean models, and Langevin-type dynamics; Bresler and Huang [42] establish sharp phase-transition results for local and message-passing style algorithms on random  $k$ -SAT using the overlap gap property.



The third arrow—**OGP**  $\Rightarrow$  **high conditional Kolmogorov complexity**  $K^t(w \mid x)$ —is therefore not a theorem and probably false in this direct generality. The viable target is narrower: show that every algorithm in a pre-registered visible class (stable, local, low-degree, DPLL-state, or front-filtration based) must cross an OGP bottleneck, and then measure the residual description length after the class template and allowed advice have been subtracted. Extending this from restricted visible classes to arbitrary uniform polynomial-time witness producers is the separate *front-shadow* or *flow-shadow* lemma; that lemma is the real Uniformity Bridge.

*Post-v1.6 guardrail ledger.* Subsequent project ledgers tested this route conservatively rather than promoting it. The 2026-06-28 OGP/low-degree literature gate records zero allowed Uniformity-Bridge rows across six source rows; the 2026-07-06 PNP-R1 front-family falsifier records zero surviving visible-front candidates across the Kizildag/MOGP and Huang–Sellke/low-degree rows. These negative ledger results do not refute the framework, but they block any claim that the current OGP/low-degree path has supplied the Uniformity Bridge. The next admissible gate is a source-bound exact front class with matched easy/planted controls plus an independent front-shadow or average-case stability lemma.

*2025/2026 literature update.* A subsequent literature pass corroborates this restricted-model pattern without altering it. Tsirkas, Wang, and Zadik [43] prove a rigorous equivalence between monotonicity of the annealed Franz–Parisi potential and low-degree MMSE lower bounds, but only for a broad family of Gaussian additive models, with no direct route to  $P$ /poly-scale circuits. Gong, Huang, Li, and Sellke [44] show, via Pitt’s correlation inequality rather than the overlap gap property, that no algorithm whose output is stable under a small Gaussian resampling of the disorder can reliably locate isolated binary-perceptron solutions; this is a pure resampling-stability lemma and, notably, does not itself invoke OGP, which broadens the guardrail’s methodological base beyond OGP/low-degree arguments alone—but it still permits unstable or brittle algorithms to succeed, so it supplies no uniform witness-producer map either. Both sources sharpen the restricted-model literature considered above; neither closes the front-shadow or Uniformity-Bridge gap, and neither is promoted to a source-bound exact-class front candidate here.

#### Threshold axiom and diagnostic framework

The Uniformity Bridge admits a precise formulation as a *threshold axiom*—the minimal structural condition that separates existential hardness (“for each  $n$ , hard instances exist”) from adversarial hardness (“a single procedure generates hard instances for all  $n$ ”).

**Axiom XI.6** (Uniformity Bridge – UB-PvNP). *Let  $\mathcal{W}_n$  denote the set of candidate high-entropy witnesses for SAT*

*instances of size  $n$ : pairs  $(\varphi_n, w)$  with  $K^t(w \mid \varphi_n) \geq n^\varepsilon$ . A near-maximal, non-uniform bridge target would assert:*

$$\begin{aligned} &\forall \text{ algorithm } A, \quad \exists^\infty n, \quad \exists \varphi_n \in \text{SAT}_n : \\ &\quad \text{all witnesses } w \text{ of } \varphi_n \\ &\quad \text{satisfy } K^t(w \mid \varphi_n) \geq n^\varepsilon. \end{aligned}$$

(Note the quantifier  $\exists^\infty n$  (“infinitely often”), not  $\forall n$  (“almost everywhere”). Theorem IV.3 proves only the weaker logarithmic-threshold version for each fixed  $(q, C)$ ; the displayed  $n^\varepsilon$  threshold is a bridge target.)

The **Uniformity Bridge** demands a uniform construction: a single deterministic procedure  $\mathcal{U}$  such that, given  $1^n$ ,  $\mathcal{U}(1^n)$  produces an instance  $\varphi_n$  with the above property for infinitely many  $n$ , using resources polynomial in  $n$ .

**Remark XI.7** (Core intuition). Non-uniformity is “hidden advice”: the hard instances  $\varphi_n$  exist but their identity depends on the algorithm  $A$  in a non-constructive way. The Uniformity Bridge is the attempt to *eliminate this advice*. The hardness of P vs NP lives precisely at this threshold: if UB succeeds,  $P \neq NP$  follows. If a proposed UB route fails, one controlled failure mode is an explicit *advice requirement*—a quantifiable non-uniformity barrier; other failure modes include parameter blow-up, a missing front/flow-shadow map, restricted-model scope, or an unclosed  $K^t$  transfer.

**Remark XI.8** (Correspondences around UB-PvNP). The following are candidate formulations and neighboring targets for the Uniformity Bridge:

- (i) **Uniform witness construction:** There exists a polynomial-time  $\mathcal{U}$  producing high-entropy instances  $\varphi_n = \mathcal{U}(1^n)$  for all  $n$ .
- (ii) **Uniform truth-table/circuit lower bounds:** There exists an explicit Boolean function family  $(f_n)_{n \geq 1}$ , constructible in P, with  $K^t(f_n) \geq n^\varepsilon$ ; in the direct-description accounting this rules out circuit descriptions shorter than  $n^\varepsilon$  bits, i.e. circuits of size  $o(n^\varepsilon / \log n)$  up to encoding constants.
- (iii) **Uniform pseudorandom generators:** There exists a PRG  $G : \{0, 1\}^{n^\delta} \rightarrow \{0, 1\}^n$  computable in P that fools all polynomial-size circuits.
- (iv) **Parameter-uniform separation:** The constants in the entropy gap  $K^t(w \mid \varphi_n) \geq n^\varepsilon$  can be chosen *independently* of  $n$ : no  $n$ -dependent thresholds, case distinctions, or lookup tables.

Each of (i)–(iii), if established in a sufficiently strong and uniform form, would imply  $P \neq NP$ . The arrows among them are not used here as proved equivalences; they indicate standard bridge routes through MCSP and hardness-vs-randomness, and each route carries its own hypotheses and quantitative losses.

**Remark XI.9** (The  $n$ -leak scanner: diagnostic for non-uniformity). To locate where uniformity breaks, scan the proof for these four signatures:

1. **Advice sites:** Any step of the form “choose for each  $n \dots$ ” without an effective selection rule.
2. **Lookup tables:** Implicit use of an  $n$ -specific list or encyclopedia (even if it “merely exists”).
3. **Diagonalization without uniformization:** An object defined by a pure existence argument (Cantor/König) that yields no uniform construction.
4. **Coded instance families:** Hardness shifted into a specially prepared family that already contains  $n$ -dependent advice.

In the current ESC framework, the  $n$ -leak sits in Theorem VII.4: the hard instance  $\varphi_n$  is produced by contradiction (“if all instances were easy,  $A$  would solve SAT”), yielding existence without construction. Bridge Target B3 (PRG-based families) is the most direct attempt to close this leak.

*Remark XI.10* (No-go mechanisms as hardness localization). UB-PvNP fails if and only if one of the following obstructions applies:

1. **Advice unavailability:** Every candidate uniformization implicitly requires  $a(n)$  bits of advice, and  $a(n) \geq n^\delta$  for some  $\delta > 0$ . This is the P/poly barrier: uniform hardness requires escaping non-uniform computation.
2. **Parameter explosion:** Every uniformized construction forces parameters to grow superpolynomially in  $n$ , making the “uniform” procedure effectively non-uniform.
3. **Instance-family trick:** The proof works only on an  $n$ -dependently chosen family, so the “solution” is really a non-uniformity outsourcing.

Each no-go mechanism, if proven, would constitute a *conditional impossibility theorem*: “ $P \neq NP$  cannot be proven by method  $X$  because  $X$  requires  $a(n)$  bits of advice.” Such results are barrier theorems in the spirit of Baker–Gill–Solovay and Razborov–Rudich, but targeted at the specific uniformity deficit.

*Remark XI.11* (Advice-erosion curve). The  $n$ -leak in Remark XI.9 can be turned into a quantitative diagnostic by allowing an explicit advice budget. For a samplable yes-distribution  $\mathcal{D}_n$ , define an informal residual-entropy profile

$$E_{\mathcal{D}}(n, b, T) := \inf_{A: |a_n| \leq b} H_{\text{res}}(W \mid X, A(X, a_n), T),$$

where  $A$  ranges over  $T$ -time procedures with at most  $b$  bits of  $n$ -dependent advice. The intended interpretation is:  $b = 0$  is true uniformity,  $b = O(\log n)$  encodes only size or regime information, while  $b \geq n^\delta$  represents hidden non-uniform address information.

The safe easy direction is only diagnostic: if an NP search problem has a uniform polynomial-time witness

selector, then the corresponding advice-erosion profile collapses already at  $b = 0$  and  $T = \text{poly}(n)$  for the selector-induced witness choice. The open bridge target is the opposite stress test: find SAT/MCSP/OGP families for which  $E_{\mathcal{D}}(n, \text{polylog}(n), \text{poly}(n))$  remains polynomially large. This would not by itself prove  $P \neq NP$ , but it would localize precisely how much non-uniform information is needed before the entropy obstruction erodes.

*Remark XI.12* (Search conductance). A complementary diagnostic measures transport rather than advice. For a SAT instance  $x$ , let  $\Omega_x$  be a canonical witness-shadow state space and  $S_x \subseteq \Omega_x$  the valid-witness region. A restricted search class  $\mathcal{C}$  (for example local, Markovian, DPLL-state, or flip-process models) induces distributions

$$\mu_0 \rightarrow \mu_1 \rightarrow \dots \rightarrow \mu_T$$

on  $\Omega_x$ . Informally,  $\Phi_{\mathcal{C}}(x)$  is the smallest flow capacity across a cut separating the initial mass from  $S_x$ .

For such restricted models one can aim for a standard bottleneck lemma: if  $\mu_0(S_x)$  is negligible and every separating cut has capacity  $\Phi_{\mathcal{C}}(x) \leq \exp(-n^\epsilon)$ , then no  $\text{poly}(n)$ -step process in  $\mathcal{C}$  can put constant mass on  $S_x$ . This is only a restricted-model obstruction. The genuinely hard “flow-shadow” lemma would have to show that any uniform polynomial-time witness producer for the chosen SAT family admits such a canonical flow representation without superpolynomial loss. That flow-shadow lemma is another precise form of the Uniformity Bridge.

*Remark XI.13* (Capillary-front filtration). The advice and flow diagnostics can be combined into a stricter front-based ledger. For an algorithm  $A$ , instance  $x$ , and advice  $a_n$ , consider a pre-registered sequence

$$F_0(A, x, a_n) \leq F_1(A, x, a_n) \leq \dots \leq F_T(A, x, a_n)$$

of low-complexity visible fronts in a witness-shadow space: partial assignments, DPLL-state classes, cluster certificates, resolution-width bands, or bounded proof-family templates. A capillary hit is an isolated path to a witness; a front is a low-complexity structure that reaches positive witness mass.

Two diagnostics must remain separate. The Shannon proxy is

$$H_{\text{front}}(x, t) = H_{\text{res}}(W \mid x, F_t),$$

whereas the actual resource-bounded/MDL residual for a witness  $w$  is

$$K_{\text{front}}^t(x, w, t) = K^t(w \mid x) - K(\text{BPF}) - K(\text{template}) - |a_n| - K(r \mid x),$$

where  $r$  denotes any permitted specialization parameter of the visible front. The first quantity is experimental evidence only; a uniformity obstruction requires the second quantity to remain large after the template, advice, and specialization bits have been debited.

For a restricted class  $\mathcal{C}$ , a possible lemma would say: if every allowed front in  $\text{poly}(n)$  time either meets only

$o(1)$  mass of  $S_x$  or leaves  $K_{\text{front}}^t(x, w, t) \geq n^\varepsilon$  on the residual witnesses, then  $\mathcal{C}$  cannot solve the family with constant success probability. This is still a restricted-model statement. To use it against all polynomial-time algorithms one would have to prove a front-shadow lemma showing that every uniform witness producer induces such a low-complexity visible front without superpolynomial loss.

The leakage ledger must also record where the front came from. A front extracted from the same solver run whose witness mass it explains is not an external uniformity signal unless the extraction rule is fully debited as template or advice length. Safe accounting gives the elementary bound

$$K^{q_T}(w \mid x) \leq K(\mathcal{C}_n) + |a_n| + K(r \mid x) + O(\log n),$$

for any pre-registered visible producer class  $\mathcal{C}_n$  with advice  $a_n$ , specialization parameter  $r$ , and runtime  $T(n)$ . A positive  $K^t$ -residue therefore excludes only the debited visible class until a separate front-shadow or flow-shadow lemma transfers the accounting to arbitrary uniform polynomial-time witness producers.

**Lemma XI.14** (UB implies separation). *If UB-PvNP (Axiom XI.6) holds with a time bound dominating any fixed polynomial-time witness selector, then:*

- (a)  $P \neq \text{NP}$  (immediate from the existence of uniform hard instances).
- (b) the explicit family produced by  $\mathcal{U}$  has the advertised uniform witness-entropy obstruction:

$$K^t(w \mid \varphi_n) \geq n^\varepsilon$$

for all witnesses of  $\varphi_n$  on infinitely many  $n$ .

UB-PvNP alone does not prove one-way functions or a global slice-entropy lower bound for SAT truth tables. Those are separate bridge targets: OWFs require an additional PRG/average-case- $K^t$  route (for example via Liu–Pass or suitable hardness-vs-randomness hypotheses), while slice-entropy lower bounds require a truth-table or circuit-lower-bound transfer. Conversely,  $P \neq \text{NP}$  implies UB-PvNP if and only if the hard instances can be made explicit—which is exactly the content of the Uniformity Bridge.

### E. The meta-pattern

Across all the problems in this research program, the structure is:

1. **Identify the positive form** (height, HR-form, capacity, entropy).
2. **Show one direction** (algebraic  $\Rightarrow$  positive,  $P \Rightarrow$  low entropy).

3. **Show the hard direction fails for obstructions** (non-algebraic violates positivity, NP-hard has high entropy).

4. **Close the loop with a composition/coupling theorem.**

For  $P$  vs  $\text{NP}$ , steps 1–3 are established in this paper. Step 4 – the uniformity bridge – remains the central challenge.

### F. Open directions

*Remark XI.15* (Levin’s universal search and uniformity collapse). Levin’s universal search algorithm  $U$  runs all programs of length  $\leq l$  in dovetailed fashion, allocating time  $2^{-|p|} \cdot T$  to program  $p$ . If  $P = \text{NP}$ , then  $U$  finds witnesses in polynomial time (with a large constant). The near-maximal witness-entropy conjecture provides a complementary perspective: if the relevant witnesses satisfy  $K^t(w \mid \varphi) \geq |w| - O(\log n)$ , then  $U$ ’s time allocation is dominated by the *correct* program, whose length is essentially  $|w|$ . The resulting runtime is  $2^{|w|} \cdot \text{poly}(n)$ —exponential. This is a heuristic optimality picture for Levin search under the stronger near-maximal hypothesis, not a consequence of Theorem IV.3.

*Remark XI.16* (Valiant–Vazirani reduction and few-witness amplification). The Valiant–Vazirani theorem reduces SAT to Unique-SAT with high probability via random hash functions. If the near-maximal witness-entropy conjecture holds for the resulting unique-witness instances, then for a random hash  $h$  the unique witness  $w_h$  of  $\varphi \wedge h(w) = 0^k$  satisfies  $K^t(w_h \mid \varphi, h) \geq |w_h| - O(\log n)$  with high probability (over  $h$ ). This “few-witness amplification” would transfer the stronger near-maximal variant to generic SAT instances, at the cost of a randomized reduction. Derandomization via Nisan–Wigderson generators would be an additional bridge target.

*Remark XI.17* (Min-entropy variant). The near-maximal  $K^t$  formulation can be relaxed to a distributional version using min-entropy: for a samplable distribution  $\mathcal{D}$  over SAT instances, define  $H_\infty^t(\mathcal{D}) = \min_{(\varphi, w) \in \text{supp}(\mathcal{D})} K^t(w \mid \varphi)$ . If  $H_\infty^t(\mathcal{D}) \geq |w| - O(\log n)$  for some samplable  $\mathcal{D}$ , then average-case hardness of NP follows (Impagliazzo’s “Pessiland”). This relaxation is a distributional bridge target for the stronger near-maximal variant and connects to the rich theory of average-case complexity and one-way functions.

*Remark XI.18* (Adversarial witness family). A stronger variant of the ESC constructs an explicit *adversarial* witness family  $\{w_n\}$  with a global consistency check: the witnesses are not independently hard but satisfy a collective constraint  $\mathcal{C}(w_1, \dots, w_N) = 1$  that is easy to verify but hard to satisfy. Any algorithm that computes  $w_n$  for all  $n \leq N$  must “know” the global structure  $\mathcal{C}$ , requiring description length  $\Omega(N)$ . This approach avoids the single-instance limitations of Corollary IV.5 and is related to the theory of interactive proofs and PCP.

*Remark XI.19* (Compression-to-circuit transfer). For restricted circuit classes  $\mathcal{C}$  (e.g.,  $\text{AC}^0$ , monotone circuits), the compression-to-circuit transfer takes a sharper form: if  $\varphi_n$  has no  $\mathcal{C}$ -circuit of size  $s$  computing its witness, then  $K_{\mathcal{C}}^t(w|\varphi_n) \geq \log \binom{2^n}{s}$  where  $K_{\mathcal{C}}^t$  denotes the  $\mathcal{C}$ -restricted Kolmogorov complexity. For  $\text{AC}^0$ , Razborov–Smolensky lower bounds give  $s \geq 2^{n^{\Omega(1)}}$ , yielding superpolynomial  $K_{\text{AC}^0}^t$  unconditionally. This provides a “proof of concept” for Bridge Target 1 in a restricted model.

*Remark XI.20* (MCSP internalization route). Allender and Das (2017) showed that MCSP (Minimum Circuit Size Problem) is hard for SZK under randomized reductions. Liu and Pass [45] proved that the average-case hardness of  $K^t$  (a close relative of MCSP) is *equivalent* to the existence of one-way functions. Combined with Hirahara’s [27] non-black-box worst-case-to-average-case reductions within NP, this provides a route from MCSP hardness to OWF existence. Our framework extends this chain:

$$\begin{aligned} \text{MCSP hard} &\xrightarrow{\text{Liu–Pass}} \text{OWF exist} \\ &\xrightarrow{\text{Imp.–Levin}} \text{avg-case NP-hard} \\ &\Rightarrow \text{P} \neq \text{NP}. \end{aligned}$$

The first implication uses the Liu–Pass equivalence between  $K^t$  average-case hardness and OWF existence (the connection to MCSP follows from the Allender et al. correspondence between MCSP and  $K^t$ ); the second is a standard reduction; the third is immediate (average-case NP-hardness trivially implies  $\text{P} \neq \text{NP}$ , since  $\text{P} = \text{NP}$  would make NP easy even on average). More precisely, this route requires the Liu–Pass distributional setting for a samplable  $K^t$ /MCSP-like problem, the relevant randomized or non-black-box reductions, and the appropriate success metric; it is not a generic claim that worst-case MCSP hardness alone implies  $\text{P} \neq \text{NP}$ . Thus, proving average-case hardness of the relevant  $K^t$ /MCSP variant under those hypotheses would, via Liu–Pass plus standard reductions, yield  $\text{P} \neq \text{NP}$ .

*Remark XI.21* (Ilango’s MCSP NP-hardness). Hirahara and Ilango [37] give evidence for MCSP hardness by proving NP-hardness under deterministic quasi-polynomial non-adaptive reductions from well-studied cryptographic and circuit lower-bound assumptions. This does not settle unconditional NP-hardness of MCSP. For the present framework, the relevance is narrower and safer: their conditional result identifies MCSP as a plausible bridge object for turning time-bounded Kolmogorov complexity into standard complexity-theoretic lower bounds.

*Remark XI.22* (Quantum extension). The entropy framework extends naturally to quantum computation. A quantum algorithm with  $T$  gates on  $n$  qubits can be described by  $O(T \log T)$  classical bits (encoding the gate sequence), so  $K(\text{desc}(U)) \leq O(T \log T)$  for the classical description of the unitary  $U$ . If  $\text{BQP} = \text{QMA}$ , then the polynomial-time quantum algorithm would replace the QMA witness, and the analogous witness entropy gap argument would

apply: the resource-bounded quantum Kolmogorov complexity of the witness would need to be simultaneously low (by the BQP algorithm) and high (for instances encoding random strings). This suggests  $\text{BQP} \neq \text{QMA}$ , though making this rigorous requires a quantum analog of resource-bounded Kolmogorov complexity, which is an active area of research.

*Remark XI.23* (Game-theoretic interaction model). The P vs NP question admits a game-theoretic formulation: an entropy maximizer (“Nature”) selects witness distributions to maximize  $K^t(w|\varphi)$ , while a circuit minimizer (“Algorithm”) selects circuits to minimize computation time. The minimax value of this zero-sum game represents the optimal trade-off between witness entropy and circuit size. A stronger near-maximal variant would assert that Nature can keep this value positive in a scale-sensitive sense for infinitely many  $\varphi$ . This formulation connects to the FST program’s use of minimax principles in other domains (Yang–Mills, turbulence), but it is not a theorem of the present paper.

*Remark XI.24* (Quantum certificates and MIP\*). The near-maximal witness-entropy variant can be translated into the language of quantum interactive proofs: an entropy gap  $K^t(w|\varphi) \geq |w| - O(\log n)$  would imply that no polynomial-size quantum circuit can prepare a quantum state  $|\psi_w\rangle$  encoding the witness from  $|\varphi\rangle$  alone. In the MIP\* framework (Natarajan–Wright [35], Ji–Natarajan–Vidick–Wright–Yuen [36]), entanglement between provers can be viewed as a mechanism for “distributing” the witness entropy across non-communicating parties. Whether entanglement can circumvent such a strengthened entropy barrier is an open analogy, not a claim of this framework.

*Remark XI.25* (Self-referential perspective – informal). A heuristic perspective on why closing the uniformity bridge is hard: for any fixed polynomial-time machine  $A$  of description length  $|A|$  and running-time polynomial  $q_A$ , Theorem IV.3 (conditional on  $\text{P} \neq \text{NP}$ ) gives, for every sufficiently large logarithmic constant  $C$ , infinitely many instances  $\varphi$  whose valid witnesses all satisfy  $K^{q_A}(w|\varphi) > C \log |\varphi|$ . Since  $A$  is a fixed program,  $K^{q_A}(A(x)|x) \leq |A| + O(\log |x|) \leq C \log |x|$  for all large  $x$ . These two bounds are incompatible on the corresponding hard instances:  $A$  cannot produce a valid witness there. This is simply the non-uniform direction of the witness entropy gap (already established in Theorem IV.3) and does *not* close the uniformity bridge, which requires hard instances that resist *all* algorithms simultaneously, not just each algorithm individually.

*Remark XI.26* (Extended Frege and MCSP). Two concrete targets for unconditional progress:

1. *Extended Frege*: If superpolynomial lower bounds on Extended Frege proof length can be established for tautologies encoding the unsatisfiability of ESC-hard instances, then the proof objects themselves have high entropy ( $K(\pi|\tau) \geq |\pi|^{1-\epsilon}$ ), realizing Bridge Target 2.

2. *MCSP and non-uniformity*: The Minimum Circuit Size Problem (Allender–Hirahara [34]) is the natural decision version of Kolmogorov complexity for circuits. MCSP lower bounds are a neighboring route for converting compression statements into circuit lower bounds, but  $\text{MCSP} \notin \text{P/poly}$  would not by itself imply EH without an additional reduction from small SAT-slice descriptions to small MCSP circuits, or a comparable uniform compression-to-circuit theorem.

*Remark XI.27* (Loose analogy with the Hodge conjecture). At a very high level, the ESC framework shares a structural pattern with the Hodge conjecture: objects that admit structured descriptions (polynomial-time algorithms / algebraic cycles) form a proper subset of a larger space, and the question is whether certain distinguished objects (NP-witnesses / Hodge classes) lie in this subset. However, this analogy is *superficial*: the Hodge–Riemann bilinear form is a deep topological invariant on cohomology, while  $K^t \geq 0$  is a trivial property of any length function. The analogy should be understood as a heuristic parallel in proof strategy, not as evidence for a mathematical connection.

### G. Post-Zookeeper transfer: Williams-normalized uniformity

The post-Zookeeper role of the present paper is to make the uniformity gap operational. The relevant recent input is Williams’ square-root-space simulation of time-bounded computation: time  $t$  can be simulated in space  $O(\sqrt{t} \log t)$  for the relevant Turing-machine model. This does not resolve P versus NP, but it changes the geometry of the resources that the entropy obstruction must survive.

Relative to the current CORE status, the Zookeeper closure of Riemann- $C2/MS2$  is a *normal-form constraint*, not a transfer proof for complexity theory. It licenses the five-slot bookkeeping below, but it does not close the uniformity bridge. The open mathematical step in this paper remains the passage from non-uniform high-entropy witnesses to a uniform lower bound that survives the best available time–space simulation.

The transfer normal form is:

**Operator.:** Uniform search and verification operators, measured by resource-bounded Kolmogorov complexity  $K^t$ .

**Defect.:** The uniformity defect: high-entropy witness existence minus uniform generation under the best available time–space simulation.

**Approximation.:** SAT slices, resolution-width regimes, MCSP/XOR/MUX candidates, and circuit-size levels.

**Gap.:** A residual entropy hardness gap after time has been normalized by the square-root-space simulation.

**Limit.:** Passage from non-uniform instance families to a uniform language or algorithm.

This suggests a refined diagnostic:

$$H_{\text{slice}}^{\text{ts}}(n; s) = \text{witness entropy after simulation} \\ \text{in space } s, \quad s \asymp \sqrt{t \log t}.$$

The next target is not a direct proof of  $\text{P} \neq \text{NP}$ , but a transfer lemma: if a uniform polynomial-time search procedure exists, then this Williams-normalized slice entropy must collapse along the associated resource surface. MCSP-style candidates and resolution-width families are then the natural stress tests for non-collapse.

### Result classification

This paper is a **framework and presentation paper**: it presents the known equivalence between the  $\text{P} \neq \text{NP}$  conjecture and an entropic condition (ESC) in a unified framework, without proving the separation itself (see Table VI for a comprehensive classification of proven equivalences, structural contributions, exploratory numerics, and open bridge targets).

## XII. CONCLUSION

The equivalence between  $\text{P} \neq \text{NP}$  and the incompressibility of NP-witnesses (under resource-bounded Kolmogorov complexity) has been known in various forms since the work of Orponen et al. [25] and Fortnow [24]. This paper presents these known connections in a unified framework, introducing the terminology of algorithmic positivity and the entropic No-Go theorem.

The witness entropy gap – the discrepancy between the high conditional complexity of witnesses (under  $\text{P} \neq \text{NP}$ ) and the low complexity that polynomial-time algorithms can produce – provides a clean characterization of what any proof of  $\text{P} \neq \text{NP}$  must establish. The barrier analysis argues that  $K$ -based arguments are not blocked by the three classical barriers, though this immunity is rigorous only for unbounded  $K$  and remains heuristic for the resource-bounded  $K^t$  used in the core results.

The remaining challenge is the uniformity bridge: converting the existential entropy bound into a universal lower bound against all polynomial-time algorithms. The bridge strategies identified (instance compression, proof complexity, PRG-based families, MCSP) are themselves major open problems, each essentially equivalent to or harder than P vs NP itself. The value of the framework lies in making this structure explicit.

*‘P vs NP does not ask whether hard problems exist. It asks whether the universe is efficient at finding answers once they exist. The entropy perspective shows: if the universe is not efficient, it is because the answers themselves carry too much information to be short.’*

Table VI. Result classification.

| Result                                                    | Type                         | Reference                  |
|-----------------------------------------------------------|------------------------------|----------------------------|
| <i>Proven equivalences:</i>                               |                              |                            |
| $\text{ESC} \Leftrightarrow \text{P} \neq \text{NP}$      | Proposition + Remark         | Prop. V.6, Rem. V.7        |
| $\text{EH} \Rightarrow \text{P} \neq \text{NP}$           | Theorem                      | Thm. VII.4                 |
| <i>Structural contributions:</i>                          |                              |                            |
| Barrier immunity (S1–S5 for $K$ ; heuristic for $K^t$ )   | Analysis                     | Sec. II, Rem. III.6        |
| Resolution-width conjecture                               | Conjecture                   | Conj. IX.2                 |
| MCSP internalization route                                | Remark                       | Rem. XI.20                 |
| Advice-erosion curve                                      | Bridge target                | Rem. XI.11                 |
| Search-flow conductance                                   | Bridge target                | Rem. XI.12                 |
| Capillary-front filtration                                | Bridge target                | Rem. XI.13                 |
| OGP/low-degree literature gate                            | Negative guardrail           | Project ledger, 2026-06-28 |
| PNP-R1 front-family falsifier                             | Negative guardrail           | Project ledger, 2026-07-06 |
| <i>Exploratory numerics (not constituting evidence):</i>  |                              |                            |
| Phase transition structure observed                       | Illustrative                 | Rem. VII.8                 |
| $H_{\text{slice}}$ growth trend ( $n \leq 16$ , 5 points) | Illustrative                 | Rem. VII.8                 |
| <i>Open bridge targets:</i>                               |                              |                            |
| Uniformity Bridge                                         | Open problem                 | Sec. XI D                  |
| Entropy Hardness (EH)                                     | Strong sufficient conjecture | Sec. VII                   |

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